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New Zealand FOODfiles™ 2018 Manual

June 2019



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Acknowledgement of data sources

Most of the data found in the FOODfiles™ 2018 are derived from analytical laboratory results from appropriately accredited laboratories. Some of the data have been obtained from overseas food composition databases/tables, Nutrition Information Panels on packaged foods, and the New Zealand Manufactured Food Database; and some data have been imputed from the related foods or derived from the recipe calculation by applying the nutrition retention factor. The following sources are acknowledged for the contribution of food composition data:

Dignan et al. (2004)
Finglas et al. (2015b)
Food Standards Australia New Zealand (2019)
Health Canada (2015)
National Institute for Health and Welfare. (2010)
U.S. Department of Agriculture (2018)
Wu Leung (1968)

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Purpose of this manual

This reference manual has been produced by The New Zealand Institute for Plant and Food Research Limited (Plant & Food Research) to facilitate use of the New Zealand FOODfiles™ 2018 Version 01, The Concise New Zealand Food Composition Tables, 13th Edition 2018 and Online Search. The manual provides technical information for users and information for computing personnel involved in setting up database management systems for users of food composition data in the form of the New Zealand FOODfiles. It also provides information regarding programmers' application software.

Preface

The New Zealand FOODfiles was first released in 1988 by the Biotechnology Division, Department of Scientific and Industrial Research (DSIR), with the majority of the data coming from overseas sources. Since then 21 upgrades have been released, with the latest being this current edition, New Zealand FOODfiles 2018 Version 01. The data have been regularly disseminated from the New Zealand Food Composition Database (NZFCD) by producing updated freely accessible versions of FOODfiles and associated products on the website www.foodcomposition.co.nz since 2011. In June 2018 the website was refreshed with new online search functions incorporated, so making data more accessible and user-friendly. The download function for the New Zealand FOODfiles 2018 and The Concise New Zealand Food Composition Tables 13th Edition 2018 remains freely accessible through the website www.foodcomposition.co.nz for all users. The 'NIP Nutrition Information Panel' facility that used to be available on the website has been integrated into the 'Search' facility as the default report option. Previous versions of New Zealand FOODfiles including New Zealand FOODfiles 2016 Version 01 are archived on the same website.

Plant & Food Research and the Ministry of Health jointly own the intellectual property in the NZFCD. The Ministry of Health is the principal funder of the database, with Plant & Food Research investing in the database management system and related infrastructure. The Ministry of Health contracts Plant & Food Research to maintain and develop the NZFCD and to disseminate data, via updated releases of New Zealand FOODfiles and The Concise New Zealand Food Composition Tables.

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1 INTRODUCTION

The New Zealand FOODfiles 2018 Version 01 (FOODfiles 2018 V 01) is the major source of verified food composition data available in New Zealand. It replaces the New Zealand FOODfiles 2016 Version 01 (FOODfiles 2016 V 01) released in May 2017.

There are two versions of the data files in the New Zealand FOODfiles 2018 V 01:

1. A standard version that contains information for 86 components of more than 2700 foods. Some foods do not have data for all 86 components because the number of core components has changed over time.
2. An unabridged version that contains information of up to 363 food components of the same (over 2700) foods listed in the standard version. Most foods do not have data for all 363 components, for a variety of reasons.

The NZFCD data files are intended to be used in database or spreadsheet programmes and, depending on their intended use, may require additional computer programming input to be used fully and effectively.

1.1 Updates from 2016 to 2018

The following updates were made to FOODfiles 2018 V 01. Further details can be found in the FOODfiles 2018 V 01 download – Update files (available via www.foodcomposition.co.nz).

- 203 Food Records (individual food or beverage allocated a unique identifier and Food Name) have been updated (Table 1). The list of the updated Food Records can be found in the update files 'New Food Records replacing the old Food Records in latest version of FOODfiles.xlsx'.
- 67 Food Records reported in the FOODfiles 2016 V 01 have been archived ('Food Records archived from latest version of FOODfiles.xlsx').
- 72 food composition data points have been updated ('Data added to or updated in the Food Records in the latest version of FOODfiles.xlsx').
- Three new food components were added to the standard version and unabridged files: 1) Vitamin A retinol activity equivalents, 2) Added sugar and 3) Free sugar.

As raw ingredients can vary slightly within and between seasons, users should note that the data contained in FOODfiles 2018 V 01 do not represent absolute values. Rather, the data provided reflect the typical composition of foods as available and/or consumed in New Zealand.

For additional information on prioritising foods for analysing or derived from recipe methods, images, sampling, analytical and nutrient plans, please contact staff at Plant & Food Research: foodcomp@plantandfood.co.nz. Some of this information can also be found in the recent publication Sivakumaran et al. (2018).

Table 1. Summary of the updated Food Records in FOODfiles™ 2018 V 01 ^a.

FOODfiles 2018	Derivation	Food Records	Type of Foods
New Foods	By analysis	92	Bakery products: cakes (chocolate, fruit and carrot), breads (French, pizza bases) and doughnut (chocolate iced, non-chocolate iced, and cream and jam filled); Alcoholic beverages: Mixed drinks (bourbon & cola, vodka & non-cola flavoured soft drink and vodka & flavoured milk); Beverages, non-alcoholic: oat milk, almond milk, almond and coconut milk blend, coconut milk, coconut water, reduced sugar juice concentrates (blackcurrant and lemon or lime) and soft drink ginger beer; Breakfast cereals: oats quick sachets (plain and flavoured); Cereals and pseudo-cereals: fried chow mein noodle, rice (brown), flours (green and yellow pea, almond, and coconut); Dairy: flavoured milks (chocolate and coffee); custard, sheep milk, Fast foods: salad (coleslaw), hash brown (potato); Fin fishes: canned fishes (mackerel and sardines); Fruits: pears (green to yellow and golden russet), nashi pear and orange (New Zealand); Meats: beef mince (prime and standard); Meat products: burger patties (beef and chicken); Nuts and seeds: coconut yoghurt (from coconut cream and coconut milk) Sugars, confectionaries and sweet spreads: honey (manuka); Vegetables and pulses: tomato, frozen vegetables (cauliflower, broccoli, green bean, butter bean, carrot, capsicum, sugar snap pea, onion, kale, spinach, edamame/soy bean, and mixed vegetables) and pulses (split chickpea).
	Partial analysis & recipe calculation	61	Bakery products: breads (ciabatta, French); Alcoholic beverages: mixed drinks (bourbon & cola, vodka & non-cola flavoured soft drink, vodka & energy drink and gin & tonic water); Breakfast cereals: porridge prepared with milk and water; Fast foods: hash brown (potato); Meat products: cooked burger patties (beef and chicken); Nuts and seeds: seed and nut mixes; Recipes: bolognaise sauce and meal, custard pudding, banana cake, chilli corn carne and meal and ratatouille; Vegetables and pulses: cooked mushroom, green banana, snow pea and cooked frozen vegetables (cauliflower, broccoli, green bean, butter bean, carrot, capsicum, sugar snap pea, onion, kale, spinach, edamame/soy bean, and mixed vegetables).
Reanalysed to replace old Food Records	By analysis	41	Bakery products: breads (ciabatta, pizza bases) and doughnut (with cinnamon and sugar dusted); Beverages, non-alcoholic: energy food drink (Milo™), juice concentrate (blackcurrant and lemon or lime) and soft drink ginger beer; Breakfast cereals: oats quick sachets (plain and flavoured); Cereals and pseudo-cereals: oats (rolled and wholegrain), rice (white and brown), flours (white rice and chickpea (besan)); Fin fishes: canned fishes (mackerel and sardines); Fruits: persimmon, date (dried), pears (green to yellow and golden russet), nashi pear, nectarine, peach, apricot, orange and mandarin; Miscellaneous: herb (coriander); Nuts and seeds: seeds (chia, linseed or flaxseed, sunflower and pumpkin), blanched almond and coconut yoghurt; Sugars, confectionaries and sweet spreads: honey (multifloral); Vegetables and pulses: mushroom, green banana, snow pea, tomato, frozen mixed vegetables and pulses (split pea).
	Partial analysis & recipe calculation	09	Breakfast cereals: porridge prepared (with standard milk and water); Cereals and pseudo-cereals: cooked rice (white and brown); Recipes: Vegetables and pulses: cooked mushroom, green banana, mixed vegetables and cooked pulses (split pea).
Total		203	

^a More details can be found in the Update files 'New Food Records replacing the old Food Records in latest version of FOODfiles.xlsx'.

2 NOTES FOR USERS

2.1 Limitations in the dataset

FOODfiles 2018 V 01 contains a listing of 2767 Food Records from the New Zealand Food Composition Database (NZFCD). Although there are 363 individual food components (nutrients and non-nutrients) in the NZFCD, only 86 of these ("core components", the most commonly referenced food components) are included in the standard version of FOODfiles 2018 V 01. Some foods do not have data for all 86 components because the set of core components has changed over time.

The unabridged version of FOODfiles 2018 V 01 includes data for up to 363 components. Total number of mean values are approximately 471,000. Where the data are unavailable for particular components, they are recorded as a missing value.

The Concise New Zealand Food Composition Tables, 13th Edition 2018 (Concise Table 13th Edition) provides a subset, or abridged version, of FOODfiles 2018 V 01 containing 1187 foods with 38 food components. It should be noted that in the Concise Table 13th Edition, food component data are expressed in weight (g) of food using the Common Standard Measure (CSM) or as commonly purchased or eaten, whereas food component data are expressed as per 100 g food or as specified for FOODfiles.

2.2 FoodID

Each food is uniquely identified by an alpha-numeric FoodID: a single letter denoting a major food chapter (see Table 2) followed by one to four digits. The FoodID is the record key that links data between NAME.FT, DATA.FT, DATA.AP, CSM.FT and INGREDIENT.FT files described in Section 3.

Table 2. Food chapters.

Chapter	Food chapter	Includes information
A	Bakery products	Breads, biscuits, muffins, pizza bases and cakes
B	Beverages, alcoholic	Beers, port, sherry, spirits and wines
C	Beverages, non-alcoholic	Drink concentrates and powders, energy drinks, fruit drinks, fruit juices, smoothie, liquid breakfast, rice milks, soy milks, teas, coffees and soft drinks
D	Breakfast cereals	Extruded cereals, mixed grain cereals and mueslis
E	Cereals and pseudo-cereals	Cereal constituents, flours, noodles, pastas, rice and tempeh
F	Dairy	Butters, cheeses, ice creams, milks, protein powder and yoghurts
G	Eggs	Chicken eggs – cooked and raw
H	Fast foods	Burgers, chain store takeaways, Chinese takeaways, fries, Indian takeaways, pizzas, savoury items and Thai takeaways
J	Fats and oils	Cooking and salad oils, margarines and shortenings
K	Fin fishes	Fresh and canned fishes
L	Fruits	Fruits raw, dried, canned and stewed

Chapter	Food chapter	Includes information
M	Meats	Variety of meat cuts and types from beef, chicken, lamb, pork, venison, etc. raw and cooked
N	Meat products	Cured meats and sausages
P	Miscellaneous	Salts, stocks, yeast and yeast spreads; spices and herbs
Q	Nuts and seeds	Nuts, seeds and their products
R	Recipes	Multiple ingredients recipes, derived from recipe calculation using weight, yield and nutrient retention factors
S	Sauces	Dressings, dips, tomato sauces, simmer sauces and soy sauces
T	Shellfishes	Cockles, crab, mussels, prawns, scallops and squid
U	Snack foods	Chips, muesli bars and popcorns
V	Soups	Soup dry mix and soup prepared
W	Sugars, confectionaries and sweet spreads	Chocolates, jams, lollies, Nutella, marmalade and sugars
X	Vegetables and pulses	Raw and cooked

As new Food Records are added to the NZFCD, they are given a unique FoodID. When Food Records are removed from the NZFCD (archived), their FoodID is also archived, i.e. never used again. This ensures that no confusion arises when new versions of the New Zealand FOODfiles are released.

2.3 Source code

All components in the New Zealand Food Composition Database are assigned a source code. This is a lower-case letter code that indicates the source of data, such as food composition analysis, calculation, or borrowed from another database. The full list of source codes is shown in Appendix 2. The common source codes are:

z	New Zealand analytical data
c	Calculated by FIMS (Food Information Management System) after data entry
p	Presumed zero
r	Value imputed from a related food
u	United States Department of Agriculture (USDA) National Nutrient Database sourced data
a	Australian Food Composition Database (previously called NUTTAB) sourced data
d	Derived from published sources.

The majority of values in FOODfiles are New Zealand analytical data (i.e. source code 'z'). Foods and nutrients prioritised for analyses are those that are frequently consumed and/or make an important contribution to nutrient intake in New Zealand. Sometimes it is not possible or necessary to analyse all components in a food. In these circumstances values are borrowed from other sources, imputed, presumed zero or calculated.

- Presumed zero: Data are presumed to be zero if the food is not a source of the particular food component. For example, for pasta, the alcohol and caffeine values are presumed zero because none of the ingredients used for preparation of pasta is a source of alcohol or caffeine.
- Borrowed data: Data are sourced from published food composition databases or food composition tables providing data of the necessary quality, provided that these data are expected to be representative of the New Zealand food.
- Imputed data: Data are sourced from similar or related foods.
- Calculated: Data can be calculated using the recipe or attribute calculator in a PFR-developed Java application, Food Information Management System (FIMS). The calculated dataset can have various combinations of source codes. For example, the source codes “cdz” for protein represent: data point by calculation (“c”) applying a conversion factor from published source “d” to New Zealand analytical value; (“z”) for nitrogen, total.

2.4 Food components

Each food component is uniquely identified by an INFOODS (International Network of Food Data Systems) tagname. Tagnames are compiled by an expert committee under the auspices of INFOODS, and allocated to unique food components (Klensin et al. 1989; FAO/INFOODS 2012c). The INFOODS tagnames incorporate the component entity, the method of analysis where this is specific to the result (e.g. dietary fibre methodologies), and the default unit of measure. As INFOODS tag names are not available for some of the food components recorded in the NZFCD, the NZFCD uses ‘Component Identifiers’. The ‘Component Identifier’ includes the available INFOODS tagname or, if there is no tag name, is generated following a similar naming convention to the INFOODS tagnames (FAO/INFOODS 2012c; Klensin et al. 1989). A list of the food components, their ‘Component Identifiers’ INFOODS tagnames, units of measure and a brief description are given in Appendix 1.

All these components’ data are derived by chemical analysis or are calculated from the chemical analytical data. Which of the components are analysed (A) and which are calculated (C) are listed in the Appendix 1. The chemical methods used for analysis of the components are summarised in the Appendix 3. The components data that are calculated from the analytical data are described in the following sections.

2.5 Proximate, total

Proximate, total is calculated by summing the following components: Ash (ASH), protein (PROT); fat, total (FAT); water (WATER); dietary fibre (FIBTG); alcohol (ALC); and available carbohydrate by weight (CHOAVL). The sum of proximates should fall within a range of 95–105 g per 100 g edible portion. A margin of plus or minus 5% is considered acceptable (FAO/INFOODS 2012c) because many of the food components are determined independently on different samples in different laboratories. In FOODfiles 2018 V 01, we accept that 8.6% of the Food Records have a sum outside this acceptable range. Explanations for this include the presence of other constituents other than the major proximates listed above, and other miscellaneous organic compounds (e.g. resistant starch, oligosaccharides and polydextrose), which are not measured independently but may contribute to the values. When the proximate analyses are outside the 5% allowance, the typical food component values are still reliable.

2.6 Energy

The energy values are calculated in four different ways according to the FAO/INFOODS guide lines (FAO/INFOODS 2012a) and Standard 1.2.8 (Food Standards Australia New Zealand 2018). They are expressed in two ways, either as kilocalories (kcal) or kilojoules (kJ), so providing eight calculated component values for energy for each food.

1. Energy, total metabolisable (kJ, ENERC)
2. Energy, total metabolisable (kcal, ENERC_KCAL)
3. Energy, total metabolisable (kJ, including dietary fibre, ENERC1)
4. Energy, total metabolisable (kcal, including dietary fibre, ENERC1_KCAL)
5. Energy, total metabolisable, carbohydrate by difference, FSANZ (kJ, ENERC_FSANZ1)
6. Energy, total metabolisable, carbohydrate by difference, FSANZ (kcal, ENERC_FSANZ1_KCAL)
7. Energy, total metabolisable, available carbohydrates, FSANZ (kJ, ENERC_FSANZ2)
8. Energy, total metabolisable, available carbohydrates, FSANZ (kcal, ENERC_FSANZ2_KCAL).

The values are calculated from the energy-producing food components (carbohydrate, protein, fat, total, and alcohol), with and without dietary fibre and other energy producing food components, using the appropriate energy conversion factors (Table 3) for FOODfiles 2018 V 01.

Table 3. Conversion factors used for calculation of energy.

Component identifier for conversion factor	Component (Component Identifier)	kJ/g	kcal/g
XPROT XPROT_KCAL	Protein, total (PROT)	17	4
XFAT XFAT_KCAL	Fat, total (FAT)	37	9
XCHOAVL XCHOAVL_KCAL	Available carbohydrate (by weight, difference (CHOAVL)) ^a	17	4
XCHOCDF XCHOCDF_KCAL	Total carbohydrate by difference (CHOCHDF)	17	4
XALC XALC_KCAL	Alcohol (ALC)	29	7
XFIBTG XFIBTG_KCAL	Dietary fibre (FIBTG)	8	2
XOA XOA_KCAL	Organic acid (OA_G)	13	3

^a The conversion factors for available carbohydrates by weight (XCHOAVL and XCHOAVL_KCAL) are used as a conversion factor for all other carbohydrates converting to energy.

Source: Food and Agriculture Organisation of the United Nations and the World Health Organisation (2003); (Food Standards Australia New Zealand 2018).

2.6.1 FAO/INFOODS (FAO/INFOODS 2012a)

1. **Energy, total metabolisable (kJ, ENERC)** is calculated from the values of energy-producing food components and appropriate conversion factors listed in Table 3. The energy-producing food components taken into account are: protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC).

Equation 01:

$$\begin{aligned} \text{ENERC (kJ/100 g)} \\ &= [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] + [\text{CHOAVL(g/100 g)} \times \text{XCHOAVL(kJ/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] + [\text{ALC(g/100 g)} \times \text{XALC (kJ/g)}] \end{aligned}$$

Where the 'Component Identifier' XPROT, XCHOAVL, XFAT and XALC are the energy conversion factors expressed in kJ/g for protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC) as listed in Table 3.

2. **Energy, total metabolisable (kcal, ENERC_KCAL)** is calculated from the energy-producing food components: protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC).

Equation 02:

$$\begin{aligned} \text{ENERC (kcal/100 g)} \\ &= [\text{PROT (g/100 g)} \times \text{XPROT_KCAL (kcal/g)}] + [\text{CHOAVL(g/100 g)} \\ &\times \text{XCHOAVL_KCAL (kcal/g)}] + [\text{FAT(g/100 g)} \times \text{XFAT_KCAL (kcal/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC_KCAL (kcal/g)}] \end{aligned}$$

Where XPROT_KCAL, XCHOAVL_KCAL, XFAT_KCAL and XALC_KCAL are the energy conversion factors expressed in kcal/g for protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), and alcohol (ALC) as listed in Table 3.

3. **Energy, total metabolisable (kJ, including dietary fibre, ENERC1)** is calculated from the energy-producing food components: protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), alcohol (ALC) and fibre, total dietary (FIBTG).

Equation 03:

$$\begin{aligned} \text{ENERC1 (kJ/100 g)} \\ &= [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] + [\text{CHOAVL(g/100 g)} \times \text{XCHOAVL(kJ/g)}] \\ &+ [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] + [\text{ALC(g/100 g)} \times \text{XALC (kJ/g)}] \\ &+ [\text{FIBTG(g/100 g)} \times \text{XFIBTG(kJ/g)}] \end{aligned}$$

Where the 'Component Identifier' XFIBTG is the energy conversion factor expressed in kJ/g for fibre, total dietary and other energy conversion factors are explained above in Equation 01.

4. **Energy, total metabolisable (kcal, including dietary fibre, ENERC1_KCAL)** is calculated from the energy-producing food components: protein (PROT), available carbohydrates by weight (CHOAVL), fat, total (FAT), alcohol (ALC) and dietary fibre (FIBTG).

Equation 04:

$$\begin{aligned} \text{ENERC1 (kcal/100 g)} &= [\text{PROT (g/100 g)} \times \text{XPROT_KCAL (kcal/g)}] + [\text{CHOAVL(g/100 g)} \\ &\times \text{XCHOAVL_KCAL (kcal/g)}] + [\text{FAT(g/100 g)} \times \text{XFAT_KCAL (kcal/g)}] \\ &+ [\text{ALC(g/100 g)} \times \text{XALC_KCAL (kcal/g)}] + [\text{FIBTG(g/100 g)} \\ &\times \text{XFIBTG_KCAL (kcal/g)}] \end{aligned}$$

Where the 'Component Identifier' XFIBTG_KCAL is the energy conversion factor expressed in kcal/g for fibre, total dietary and other energy conversion factors are explained above in Equation 02.

2.6.2 Standard 1.2.8 (Food Standards Australia New Zealand 2018)

5. **Energy, total metabolisable, carbohydrate by difference, FSANZ (kJ, ENERC_FSANZ1)** value is calculated according to Standard 1.2.8. The energy-producing food components accounted are: protein (PROT), carbohydrates by difference (CHOAVDF_FSANZ), fat, total (FAT), alcohol (ALC), dietary fibre (FIBTG) and organic acids (OA_G). The calculation of carbohydrate by difference (CHOAVDF_FSANZ) value is explained below in Section 2.9.2.

Equation 05:

$$\begin{aligned} \text{ENERC_FSANZ1 (kJ/100 g)} &= [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] + [\text{CHOAVDF_FSANZ (g/100 g)} \\ &\times \text{XCHOAVL(kJ/g)}] + [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] + [\text{ALC(g/100 g)} \\ &\times \text{XALC (kJ/g)}] + [\text{FIBTG(g/100 g)} \times \text{XFIBTG(kJ/g)}] + [\text{OA_G(g/100 g)} \\ &\times \text{XOA(kJ/g)}] \end{aligned}$$

Note: Food components 'unavailable carbohydrates', 'polyols' and 'polydextrose' are not recorded in FOODfiles 2018 V 01. Therefore they are not included when calculating for **ENERC_FSANZ1**.

6. **Energy, total metabolisable, carbohydrate by difference, FSANZ (kcal, ENERC_FSANZ1_KCAL)** value is converted from ENERC_FSANZ1 by using the unit conversion factor 4.18.

Equation 06:

$$\text{ENERC_FSANZ1_KCAL (kcal/100 g)} = \text{ENERC_FSANZ1 (kJ/100 g)} / 4.18$$

7. **Energy, total metabolisable, available carbohydrates, FSANZ (kJ, ENERC_FSANZ2)** value is calculated according to Standard 1.2.8. The energy-producing food components accounted are: protein (PROT), available carbohydrates (CHOAVL_FSANZ), fat, total (FAT), alcohol (ALC), dietary fibre (FIBTG) and organic acids (OA_G). The calculation of available carbohydrates (CHOAV_FSANZ) value is explained in Section 2.9.2.

Equation 07:

$$\begin{aligned} \text{ENERC_FSANZ2 (kJ/100 g)} \\ = [\text{PROT (g/100 g)} \times \text{XPROT(kJ/g)}] + [\text{CHOAVL_FSANZ (g/100 g)} \\ \times \text{XCHOAVL(kJ/g)}] + [\text{FAT(g/100 g)} \times \text{XFAT(kJ/g)}] + [\text{ALC(g/100 g)} \\ \times \text{XALC (kJ/g)}] + [\text{FIBTG(g/100 g)} \times \text{XFIBTG(kJ/g)}] + [\text{OA_G(g/100 g)} \\ \times \text{XOA(kJ/g)}] \end{aligned}$$

Note: Food components 'unavailable carbohydrates', 'polyols' and 'polydextrose' are not recorded in FOODfiles 2018 V 01. Therefore, they are not included when calculating for ENERC_FSANZ2.

8. **Energy, total metabolisable, available carbohydrates, FSANZ (kcal, ENERC_FSANZ2_KCAL)** value is converted from ENERC_FSANZ2 by using the unit conversion factor 4.18.

Equation 08:

$$\text{ENERC_FSANZ2_KCAL (kcal/100 g)} = \text{ENERC_FSANZ2 (kJ/100 g)} / 4.18$$

Note: Examples showing calculation of the different methods of determining energy can be found in Appendix 4.

2.7 Protein, total

For all foods protein (protein, total; calculated from total nitrogen (PROT)) is measured as total nitrogen multiplied by a specific nitrogen-to-protein conversion factor (XN) (Greenfield & Southgate 2003). The conversion factor that is used for calculation of the PROT for each food is tabulated in the FOODfiles 2018 V 01, unabridged data files.

Equation 09:

$$\text{PROT (g / 100 g food)} = \text{NT (g / 100 g food)} \times \text{XN (g / g Nitrogen)}$$

2.8 Sugars

Sugars total (SUGAR) is the sum of the free individual monosaccharides: glucose (GLUS) and fructose (FRUS), and disaccharides: lactose (LACS), maltose (MALS) and sucrose (SUCS).

Equation 10:

$$\begin{aligned} \text{SUGAR (g / 100 g food)} \\ = \text{GLUS(g/100 g)} + \text{FRUS (g/100 g)} + \text{LACS(g/100 g)} + \text{MALS(g/100 g)} \\ + \text{SUCS(g/100 g)} \end{aligned}$$

Glycogen (GLYC), lactose (LACS), maltose (MALS), sucrose (SUCS), starch, total (STARCH), and oligosaccharides, total available (OLSAC) are expressed as monosaccharide equivalents (GLYCM, LACSM, MALSM, SUCSM, STARCHM and OLSACM respectively) calculated using the conversion factors shown in Table 4.

Table 4. Conversion factors used for calculation of monosaccharide equivalents.

Food component	Conversion to monosaccharide equivalents (g/g)
Glycogen (GLYC)	x 1.10
Lactose (LACS)	x 1.05
Maltose (MALS)	x 1.05
Sucrose (SUCR)	x 1.05
Starch, total (STARCH)	x 1.10

Source: Finglas et al. (2015a)

Disaccharide, total; expressed in monosaccharide equivalents (DISACM) value is calculated by summing of lactose, maltose and sucrose expressed in monosaccharide equivalents (LACSM, MALSM and SUCSM respectively).

Equation 11:

$$\text{DISACM (g/100 g)} = \text{LACSM (g/100 g)} + \text{MALSM (g/100 g)} + \text{SUCSM (g/100 g)}$$

Sugar, total; expressed in monosaccharide equivalents (SUGARM) value is calculated by summing the free monosaccharide (GLUS and FRUS) and disaccharides expressed in monosaccharide equivalents (DISACM).

Equation 12:

$$\text{SUGARM (g/100 g)} = \text{GLUSM (g/100 g)} + \text{FRUSM (g/100 g)} + \text{DISCM (g/100 g)}$$

Added sugars (SUGAD) are defined according to the United States Food and Drug Administration (US FDA) definition: “sugars that are either added during the processing of foods, or are packaged as such, and include sugars (free, mono- and disaccharides), sugars from syrups and honey, and sugars from concentrated fruit or vegetable juices that are in excess of what would be expected from the same volume of 100 percent fruit or vegetable juices of the same type. The definition excludes fruit or vegetable juice concentrated from 100 percent fruit juice that is sold to consumers (e.g. frozen 100 percent fruit juice concentrate) as well as some sugars found in fruit and vegetables juices, jellies, jams, preserves, and fruit spreads” (Erickson & Slavin 2015).

Free sugars (SUGFR) are defined according to the World Health Organisation (WHO) definition: “free sugars include monosaccharides and disaccharides added to foods and beverages by the manufacturer, cook or consumer, and sugars naturally present in honey, syrups, fruit juices, and fruit juice concentrates” (World Health Organisation 2015).

Added and free sugars are estimated using the 10 steps method described by Louie et al. (2015) and Louie et al. (2016). Free sugars are estimated adapting (Louie et al. 2015) method with some modification (e.g. include fruit juice) described by (Kibblewhite et al. 2017). Appendix 5 describes the 10-step process; steps 1-6 being objective and steps 7-10 being subjective.

2.9 Carbohydrates

Carbohydrate values are expressed in different ways in FOODfiles 2018 V 01 to meet the definitions for INFOODS (FAO/INFOODS 2012b) and Standard 1.2.8 of the code (Food Standards Australia New Zealand 2018).

2.9.1 INFOODS (FAO/INFOODS 2012a)

The carbohydrate contents in foods are expressed in FOODfiles 2018 V 01 in five different ways based on the guidelines FAO/INFOODS (2012a). **Total carbohydrates** value includes fibre, total dietary (FIBTG) and **Available carbohydrates** value as 'by weight' or as 'monosaccharide equivalent', not including FIBTG value. Carbohydrate values are calculated either as the sum of the analytically analysed carbohydrate components or as the difference of 100 minus the sum of the other proximate components.

9. **Available carbohydrates by weight (CHOAVL)** is the sum of analytical values of sugar, total (SUGAR); starch (STARCH) and glycogen (GLYC).

Equation 13:

$$\text{CHOAVL}(\text{g}/100 \text{ g}) = \text{SUGAR}(\text{g}/100 \text{ g}) + \text{STARCH}(\text{g}/100 \text{ g}) + \text{GLYC}(\text{g}/100 \text{ g})$$

10. **Available carbohydrates in monosaccharide equivalent (CHOAVLM)** is the sum of analytical values of sugar, total (SUGAR), starch (STARCH) and glycogen (GLYC). It includes the residual water from hydrolysis around each monosaccharide (monosaccharide equivalents SUGARM, STARCHM and GLYCM).

Equation 14:

$$\text{CHOAVLM}(\text{g}/100 \text{ g}) = \text{SUGARM}(\text{g}/100 \text{ g}) + \text{STARCHM}(\text{g}/100 \text{ g}) + \text{GLYCM}(\text{g}/100 \text{ g})$$

11. **Available carbohydrate by difference (CHOAVDF)** is calculated by deducting the sum of percentage of water (WATER), protein, total (PROT), fat, total (FAT), alcohol (ALC), ash (ASH) and fibre, total dietary (FIBTG) from 100.

Equation 15:

$$\text{CHOAVDF}(\text{g}/100 \text{ g}) = 100 - [\text{WATER}(\text{g}/100 \text{ g}) + \text{PROT}(\text{g}/100 \text{ g}) + \text{FAT}(\text{g}/100 \text{ g}) + \text{ALC}(\text{g}/100 \text{ g}) + \text{ASH}(\text{g}/100 \text{ g}) + \text{FIBTG}(\text{g}/100 \text{ g})]$$

12. **Total carbohydrate by difference (CHOCDF)** is calculated by deducting the sum of percentage of water (WATER), protein, total (PROT), fat (FAT), alcohol (ALC) and ash (ASH) from 100.

Equation 16:

$$\text{CHOCDF}(\text{g}/100 \text{ g}) = 100 - [\text{WATER}(\text{g}/100 \text{ g}) + \text{PROT}(\text{g}/100 \text{ g}) + \text{FAT}(\text{g}/100 \text{ g}) + \text{ALC}(\text{g}/100 \text{ g}) + \text{ASH}(\text{g}/100 \text{ g})]$$

13. **Total carbohydrate by summation (CHOCSM)** is the sum of analytical values of sugar, total (SUGAR), starch (STARCH) oligosaccharides (OLSAC) and fibre, total dietary (FIBTG).

Equation 17:

$$\text{CHOCSM (g/100 g)} = \text{SUGAR(g/100 g)} + \text{STARCH(g/100 g)} + \text{OLSAC(g/100 g)} + \text{FIBTG(g/100 g)}$$

2.9.2 Standard 1.2.8 (Food Standards Australia New Zealand 2018)

For nutrient labelling, two definitions of carbohydrate are found in Standard 1.2.8 of the code, either Carbohydrate by difference or Available carbohydrate. Neither definition includes dietary fibre (fibre, total dietary (FIBTG)).

Carbohydrate by difference, FSANZ (CHOAVDF_FSANZ) is calculated by deducting the sum of percentage of water (WATER), protein (protein, total; PROT), fat (fat, total; FAT), dietary fibre (fibre, total dietary (FIBTG)), ash (ASH), alcohol (ALC) and, if qualified or added to the food (any amount as additive or ingredients to the final food), any other unavailable carbohydrates and substances: sugar; alcohols [erythritol (ERYTHL), glycerol (GLYRL), isomalt (ISOMAL), lactitol (LACTL), maltitol (MALTL), mannitol (MANTL), sorbitol (SORTL), xylitol (XYLTL)], D-tagatose, polydextrose and organic acids, total (OA_G) from 100.

Equation 18:

$$\begin{aligned} \text{CHOAVDF_FSANZ1(g/100 g)} \\ = 100 - [\text{WATER (g/100 g)} + \text{PROT(g/100 g)} + \text{FAT (g/100 g)} \\ + \text{FIBTG(g/100 g)} + \text{ASH(g/100 g)} + \text{ALC(g/100 g)} + \text{OA_G (g/100 g)}] \end{aligned}$$

Note: **Carbohydrate by difference (CHOAVDF_FSANZ)** value is equal to the **Available carbohydrate; by difference (CHOAVDF)** value if other additives or ingredients and unavailable carbohydrates and substances are not present or accounted for. The food components: erythritol, glycerol, isomalt, lactitol, maltitol, mannitol, sorbitol, xylitol, D-tagatose, and polydextrose are not recorded in FOODfiles 2018 V 01 and therefore are not included in this equation. A few foods do have values for organic acids, total (OA_G).

14. **Available carbohydrate (CHOAVL_FSANZ)** is the sum of analytical values of total available sugars (sugar, total; SUGAR) and starch (STARCH) and, if qualified or added to the food (any amount as additive or ingredients to the final food), any available oligosaccharides (oligosaccharides, total available, OLSAC), glycogen (GLYC) and maltodextrin (MALTDEX).

Equation 19:

$$\text{CHOAVL_FSANZ} = \text{SUGAR (g/100 g)} + \text{STARCH(g/100 g)} + \text{GLYC(g/100 g)} + \text{MALTDEX(g/100 g)}$$

Note: **Available carbohydrate (CHOAVL_FSANZ)** value is equal to the **Available carbohydrates by weight (CHOAVL)** value if values for oligosaccharides and maltodextrin are not present or accounted for. The food components: oligosaccharides, total available (OLSAC) are not recorded in FOODfiles 2018 V 01. A few foods do have values for glycogen (GLYC) and maltodextrin (MALTDEX).

Carbohydrate exchange is the weight (g) of food containing 10 g carbohydrate, available. It can be calculated using the following formula (Monro 1999).

Equation 20:

$$\text{Carbohydrate exchange} = \frac{\text{Carbohydrate, total (g/100 g)}}{10}$$

2.10 Fatty acids

2.10.1 Fatty acid, total (FACID) ¹

Fatty acid, total (FACID) is determined in two different ways for FOODfiles 2018: V 01

Calculated by multiplying the fat, total (FAT) with fat-to-fatty-acid conversion factor (XFA) (Klensin et al. 1989).

Equation 21:

$$\text{FACID (g/100 g food)} = \text{FAT(g/100 g food)} \times \text{XFA(g/g)}$$

XFA values are recorded in the literature (Atwater & Woods 1896);(Weihrauch et al. 1977);(Greenfield & Southgate 2003). The conversion factors used for each food are found in the unabridged data files of FOODfiles 2018 V 01.

Determined from the analytical chromatography method by Sukhija and Palmquist (1988) so a conversion factor is not required.

Note: The method for determining the fatty acid, total value is not overtly specified in the FOODfiles 2018 V01; either method can be used and they are all tabulated under the **Fatty acid, total (FACID)**. However, if XFA is listed for a food, the FACID has been calculated using Equation 21 above and if the XFA value is not listed (left as blank) then the FACID has been determined analytically.

Individual fatty acids

Fatty acids data are expressed as both g/100 g of food and as a percentage of the total fatty acid content. Individual fatty acid (g/100 g food) values in food are derived from the value of each individual fatty acid (g/100 g TFA²) using the following equation.

Equation 22:

$$\begin{aligned} \text{Individual fatty acid (g / 100 g food)} \\ = (\text{Individual fatty acid (g/100 g TFA)} \times \text{FACID (g / 100 g food)})/100 \end{aligned}$$

The individual fatty acids in the FOODfiles 2018 V 01 are listed in Tables 5–7 with the 'Component Identifier', component names systematic names, and trivial names (Klensin et al. 1989; Beare-Rogers et al. 2001; AOCS Lipid Library 2018).

¹ FACID is INFOOD tag name for Fatty acid, total; determined by calculation using equation 21.

² TFA is a commonly used abbreviation for the total fatty acid which is equivalent to the 'Component Identifier' fatty acid, total (FACID); Note, TFA is not an INFOODS tag name or 'Component Identifier'. This abbreviation TFA is used for expressing the fatty acid content (g) as per 100 g of total fatty acid (g/100 g TFA).

2.10.2 Fatty acids, total saturated (FASAT)

The **Fatty acids, total saturated (FASAT)** is the sum of all the individual saturated fatty acids listed in Table 5.

Table 5. Details of the saturated fatty acids included in FOODfiles™ 2018.

Component Identifier	Fatty acid name	Systematic name	Trivial name
F4D0	Fatty acid 4:0	butanoic	butyric
F6D0	Fatty acid 6:0	hexanoic	caproic
F8D0	Fatty acid 8:0	octanoic	caprylic
F10D0	Fatty acid 10:0	decanoic	capric
F12D0	Fatty acid 12:0	dodecanoic	lauric
F13D0	Fatty acid 13:0	tridecanoic	tridecyclic
F14D0	Fatty acid 14:0	tetradecanoic	myristic
F15D0	Fatty acid 15:0	pentadecanoic	pentadecylic acid
F16 D0	Fatty acid 16:0	hexadecanoic	palmitic
F17D0	Fatty acid 17:0	heptadecanoic	margaric
F18D0	Fatty acid 18:0	octadecanoic	stearic
F19D0	Fatty acid 19:0	nonadecanoic	nonadecylic
F20D0	Fatty acid 20:0	eicosanoic	arachidic
F21D0	Fatty acid 21:0	heneicosanoic	heneicosylic
F22D0	Fatty acid 22:0	docosanoic	behenic
F23D0	Fatty acid 23:0	tricosanoic	tricosylic
F24D0	Fatty acid 24:0	tetracosanoic	lignoceric

2.10.3 Fatty acids, total monounsaturated (FAMU)

Fatty acids, total monounsaturated (FAMU) is the sum of undifferentiated individual fatty acids listed in Table 6. The undifferentiated fatty acid, e.g. fatty acid 18:1 omega-9 (F18D1N9), is calculated from the sum of its differentiated isomers, i.e. *cis* (F18D1CN9) and *trans* (F18D1TN9).

Table 6. Details of the monounsaturated fatty acids included in FOODfiles™ 2018.

Component Identifier	Fatty acid name	Systematic name	Trivial/common name
F10D1	Fatty acid 10:1	decanoic	
F12D1	Fatty acid 12:1	dodecenoic	
F14D1	Fatty acid 14:1	tetradecenoic	myristoleic
F14D1N5	Fatty acid 14:1 omega-5	tetradec-9-enoic	myristoleic
F15D1	Fatty acid 15:1	pentadecenoic	oncobic
F16D1	Fatty acid 16:1 undifferentiated	hexadecenoic	palmitoleic
F16D1C	Fatty acid <i>cis</i> 16:1	<i>cis</i> -hexadecenoic	<i>cis</i> -palmitoleic
F16D1T	Fatty acid <i>trans</i> 16:1	<i>trans</i> -hexadecenoic	<i>trans</i> -palmitoleic
F17D1	Fatty acid 17:1	heptadecenoic	civetic
F18D1	Fatty acid 18:1 undifferentiated	octadecenoic	
F18D1C	Fatty acid <i>cis</i> 18:1	<i>cis</i> -octadecenoic	
F18D1T	Fatty acid <i>trans</i> 18:1	<i>trans</i> -octadecenoic	
F18D1N7	Fatty acid 18:1 omega-7 undifferentiated	octadec-11-enoic	vaccenic
F18D1CN7	Fatty acid <i>cis</i> .18:1 omega-7	<i>cis</i> -octadec-11-enoic	<i>cis</i> -vaccenic
F18D1TN7	Fatty acid <i>trans</i> 18:1 omega-7	<i>trans</i> -octadec-11-enoic	<i>trans</i> -vaccenic
F18D1N9	Fatty acid 18:1 omega-9 undifferentiated	octadec-9-enoic	
F18D1CN9	Fatty acid <i>cis</i> 18:1 omega-9	<i>cis</i> -octadec-9-enoic	elaidic, (<i>cis</i> -)oleic
F18D1TN9	Fatty acid <i>trans</i> 18:1 omega-9	<i>trans</i> -octadec-9-enoic	elaidic, <i>trans</i> -oleic
F20D1	Fatty acid 20:1 undifferentiated	eicosenoic	
F20D1N11	Fatty acid 20:1 omega-11	eicos-9-enoic	gadoleic
F20D1N9	Fatty acid 20:1 omega-9	eicos-11-enoic	gondoic
F22D1	Fatty acid 22:1 undifferentiated	docosenoic	
F22D1N9	Fatty acid 22:1 omega-9	docos-13-enoic	erucic
F22D1N11	Fatty acid 22:1 omega-11	docos-11-enoic	catelaidic
F24D1	Fatty acid 24:1	tetracosenoic	nervonic

2.10.4 Fatty acids, total polyunsaturated (FAPU)

Fatty acids, total polyunsaturated (FAPU) is the sum of undifferentiated individual fatty acids listed in Table 7.

Table 7. Details of the polyunsaturated fatty acids included in FOODfiles™ 2018.

Component Identifier	Fatty acid name	Systematic name	Trivial/ common name	Abbreviation
F18D2	Fatty acid 18:2 undifferentiated	octadecadienoic		
F18D2C	Fatty acid <i>cis</i> 18:2	<i>cis</i> -octadecadienoic		
F18D2T	Fatty acid <i>trans</i> 18:2	<i>trans</i> -octadecadienoic		
F18D2N6	Fatty acid 18:2 omega-6 undifferentiated			
F18D2CN6	Fatty acid <i>cis,cis</i> 18:2 omega-6	<i>cis,cis</i> -octadeca-9,12-dienoic	linoleic	
F18D2TN6	Fatty acid <i>trans</i> 18:2 omega-6			
F18D2CN9T11	Fatty acids <i>cis,trans</i> 18:2 omega-9, 11	<i>cis,trans</i> -octadeca-9,11-dienoic	rumenic	CLA
F18D3	Fatty acid 18:3 undifferentiated	octadecatrienoic	linolenic	
F18D3N3	Fatty acid 18:3 omega-3	all- <i>cis</i> -octadeca-9,12,15-trienoic	α-linolenic	ALA
F18D3TN3	Fatty acid <i>trans</i> 18:3 omega-3	octadeca-9,12,15-trienoic		
F18D3N6	Fatty acid 18:3 omega-6	all- <i>cis</i> -octadeca-6,9,12-trienoic	γ-linolenic	GLA
F18D4	Fatty acid 18:4	octadecatetraenoic	parinaric	
F18D4N3	Fatty acid 18:4 omega-3	all- <i>cis</i> -octadeca-6,9,12,15-tetraenoic	stearidonic	
F20D2	Fatty acid 20:2	eicosadienoic		
F20D2N6	Fatty acid 20:2 omega-6	all- <i>cis</i> -eicosa-11,14-dienoic	eicosadienoic	
F20D3	Fatty acid 20:3 undifferentiated	eicosatrienoic		
F20D3N3	Fatty acid 20:3 omega-3	all- <i>cis</i> -eicosa-11,14,17-trienoic	eicosatrienoic	ETE
F20D3N6	Fatty acid 20:3 omega-6	all- <i>cis</i> -eicosa-8,11,14-trienoic	dihomo-γ-linolenic	DHLA
F20D4	Fatty acid 20:4 undifferentiated	eicosatetraenoic		
F20D4N3	Fatty acid 20:4 omega-3	all- <i>cis</i> -eicosa-8,11,14,17-tetraenoic	eicosatetraenoic	ETA
F20D4N6	Fatty acid 20:4 omega-6	all- <i>cis</i> -eicosa-5,8,11,14-tetraenoic	arachidonic	AA
F20D5	Fatty acid 20:5	eicosapentaenoic		
F20D5N3	Fatty acid 20:5 omega-3	all- <i>cis</i> -eicosa-5,8,11,14,17-pentaenoic	timnodonic	EPA

Component Identifier	Fatty acid name	Systematic name	Trivial/ common name	Abbreviation
F22D2	Fatty acid 22:2	docosadienoic		
F22D2N6	Fatty acid 22:2 omega-6	all- <i>cis</i> -docosa-13,22-dienoic	docosadienoic	
F22D4	Fatty acid 22:4	docosatetraenoic		
F22D4N6	Fatty acid 22:4 omega-6	all- <i>cis</i> -docosa-7,10,13,22-tetraenoic	adrenic	
F22D5	Fatty acid 22:5 undifferentiated	docosapentaenoic		
F22D5N3	Fatty acid 22:5 omega-3	all- <i>cis</i> -docosa-7,10,13,22,19-pentaenoic	clupanodonic	DPA
F22D5N6	Fatty acid 22:5 omega-6	all- <i>cis</i> -docosa-4,7,10,13,22-pentaenoic	clupanodonic	
F22D6	Fatty acid 22:6	docosahexaenoic		
F22D6N3	Fatty acid 22:6 omega-3	all- <i>cis</i> -docosa-4,7,10,13,22,19-hexaenoic	cervonic	DHA

Fatty acids, total omega-3 polyunsaturated (FAPUN3) and **Fatty acids, total omega-6 polyunsaturated (FAPUN6)** are the sum of the undifferentiated omega-3 fatty acid and undifferentiated omega-6 fatty acid in Table 7 respectively.

2.10.5 Fatty acids, total *trans* (FATRN)

Fatty acids, total *trans* (FATRN) is the sum of the percentage of individual *trans* fatty acids in Tables 6 and 7.

The list of the fatty acids that contribute to the FATRN are summarised in Appendix 6.

2.11 Vitamins

The following vitamin data are expressed in equivalents. In the FOODfiles 2018V 01, they are calculated using appropriate conversion factors as per Table 8.

Table 8. Conversion factors used to determine equivalents of selected vitamins.

Compound Identifier	Description	Conversion factor (g/g)
Vitamin A related		
XBCARTA	Conversion factor for beta-carotene equivalent of alpha-carotene	0.5
XRECARTB XRECARTBEQ	Conversion factor for retinol equivalent of beta-carotene or beta-carotene equivalent	0.167 (1/6)
XRAECARTB XRAECARTBEQ	Conversion factor for retinol activity equivalent of beta-carotene or beta-carotene equivalent	0.083 (1/12)
Vitamin B3 related		
XTRYP	Conversion factor for niacin equivalents of tryptophan	0.017
Vitamin E related		
XTOCOPHA	Conversion factor for alpha tocopherol equivalent of alpha tocopherol	1
XTOCOPHB	Conversion factor for alpha tocopherol equivalent of beta tocopherol	0.4
XTOCOPHD	Conversion factor for alpha tocopherol equivalent of delta tocopherol	0.01
XTOCOPHG	Conversion factor for alpha tocopherol equivalent of gamma tocopherol	0.1
Dietary folate related		
XFOLAC	Conversion factor for dietary folate equivalents of folic acid	1.67

Source: Finglas et al. (2015a) and NHMRC (2006).

2.11.1 Beta-carotene equivalents (CARTBEQ)

Beta-carotene equivalents (CARTBEQ) is calculated by multiplying alpha-carotene (CARTA) by the conversion factor for beta-carotene equivalent of alpha-carotene (XCARTA) and adding beta-carotene (CARTB).

Equation 23:

$$CARTBEQ (\mu g/100 g \text{ food}) = CARTB (\mu g/100 g) + [CARTA(\mu g/100 g \text{ food}) \times XBCARTA (g/g)]$$

Note: Other provitamin A carotenoids are not reported in FOODfiles 2018 V 01, therefore are not included in beta-carotene equivalents calculation. However, for a few foods where data are “borrowed” from other sources, the food component cryptoxanthins might be included as half of the activity of beta-carotene where that is the standard practice of the original source.

2.11.2 Vitamin A, total: expressed as retinol equivalents (VITA)

Vitamin A, total: expressed as retinol equivalents (VITA) is calculated by multiplying beta-carotene equivalents (CARTBEQ) by the conversion factor (1/6, i.e. 0.167) and adding retinol (RETOL).

Equation 24:

$$VITA (\mu g/100 g \text{ food}) = RETOL(\mu g/100 g \text{ food}) + \left[\frac{CARTBEQ(\mu g/100 g \text{ food})}{6} \right]$$

2.11.3 Vitamin A, total: expressed as retinol activity equivalents (VITA_RAE)

Vitamin A, total: expressed as retinol activity equivalents (VITA_RAE) is calculated by multiplying beta-carotene equivalents (CARTBEQ) by the conversion factor (1/12, i.e. 0.083) and adding retinol (RETOL).

Equation 25:

$$VITA (\mu g/100 g \text{ food}) = RETOL(\mu g/100 g \text{ food}) + \left[\frac{CARTBEQ(\mu g/100 g \text{ food})}{12} \right]$$

2.11.4 Niacin equivalent from tryptophan (NIATRP)

Niacin equivalents, from tryptophan (NIATRP); is calculated by multiplying tryptophan (TRYP) by the conversion factor for niacin equivalents of tryptophan (XTRYP).

Equation 26:

$$NIATRP (mg/100 g \text{ food}) = TRYP(mg/100 g \text{ food}) \times XTRYP(g/g)$$

Where the tryptophan value was not available, it has been imputed as 1.1% of total protein (Health Canada 2015)

2.11.5 Niacin equivalents total (NIAEQ)

Niacin equivalents total (NIAEQ); is the sum of the percentage of niacin, preformed (NIA) and niacin equivalent from tryptophan (NIATRP).

Equation 27:

$$NIAEQ(mg/100 g) = NIA(mg/100 g) + NIATRP(mg/100 g)$$

2.11.6 Vitamin E, expressed as alpha-tocopherol equivalents (VITE)

Vitamin E, expressed as alpha-tocopherol equivalents (VITE) is calculated by multiplying alpha- beta-, gamma- and delta-tocopherols (TOCPHA, TOCPHB, TOCHPG, and TOCHPD) by their corresponding conversion factors for alpha tocopherol equivalents of XTOCPHA, XTOCPHB, XTOCHPG, and XTOCHPD respectively and summing them.

Equation 28:

$$\begin{aligned} VITE (mg/100 g food) &= [TOCPHA(mg/100 g food) \times XTOCPHA] + [TOCPHB(mg/100 g food) \\ &\times XTOCPHB] + [TOCHPG(mg/100 g food) \times XTOCHPG] \\ &+ [TOCHPD(mg/100 g food) \times XTOCHPD] \end{aligned}$$

Note: Where the tocopherol profile is incomplete, it has been assumed all activity is alpha-tocopherol. There is an exception for seeds and seed oils where the gamma-tocopherol and other forms may be present in significant amounts and part of the tocopherol calculation for VITE. Tocotrienols are not recorded in FOODfiles 2018, therefore, they are not included in the VITE calculation.

2.11.7 Dietary folate equivalents (FOLDFE)

Dietary folate equivalents (FOLDFE) is calculated by multiplying folic acid (FOLAC) by the conversion factor for dietary folate equivalents of folic acid (XFOLAC) and adding food folate, naturally occurring folates (FOLFD).

Equation 29:

$$FOLDFE (mg/(100 g food)) = [FOLAC (mg/100 g food) \times XFOLAC] + FOLFD$$

Folate food, naturally occurring food folate (FOLFD) is calculated by subtracting folic acid (FOLAC) from folate, total (FOL).

Equation 30:

$$FOLFD (mg/100 g food) = FOL (mg/100 g food) - FOLAC (mg/100 g food)$$

2.11.8 Vitamin D

Vitamin D (VITD) is the sum of cholecalciferol or vitamin D3 (CHOCAL) and ergocalciferol or vitamin D2 (ERGCAL).

Equation 31:

$$VITD (\mu g/100 g) = CHOCAL(\mu g/100 g) + ERGCAL(\mu g/100 g)$$

The amino acid (mg/g N) values can be used to calculate the amounts of amino acids in 100 g food by using the total nitrogen values for foods containing proteins with the same amino acid profiles (Orr & Watt 1957; Paul & Southgate 1978).

2.12 Amino acids

Amino acid profiles are expressed in three ways: mg/g of total nitrogen (mg/g N), mg/100 g of food and g/100 g of food. The amino acids have been determined by analysis as mg/100 g of food. The values can be expressed as mg/g N using the total nitrogen (NT) value of the food as follows:

Equation 32:

$$\text{Amino acid (mg/g N)} = \text{Amino acid (mg/100 g food)} / \text{Nitrogen total (g/100 g food)}$$

The amino acid (mg/g N) values can be used to calculate the amounts of amino acids in 100 g food by using the total nitrogen values for foods containing proteins with the same amino acid profiles (Orr & Watt 1957; Paul & Southgate 1978).

2.13 Organic acids, total

Organic acid, total (grams, OA_G) is the sum of the gram per 100 g of acetic acid (ACEAC_G), citric acid (CITAC_G), lactic acid (LACAC_G), malic acid (MALAC_G), oxalic acid (OXALAC_G) and quinic acid (QUINAC_G).

Equation 33:

$$OA_G(g/100\ g) = ACEAC_G(g/100\ g) + CITAC_G(g/100\ g) + LACAC_G(g/100\ g) + MALAC_G(g/100\ g) + OXALAC_G(g/100\ g) + QUINAC(g/100\ g)$$

Note: Only a few foods have values for the organic acids.

2.14 Recipes

Recipe Food Records are generated by calculation following a method known as 'Mixed method' (FAO/INFOODS 2012a) using FIMS. In this recipe method Nutrient Retention Factors (NRFs) are applied at the individual ingredient level and the Yield Factors (YFs) applied at the recipe level. The calculations range from simple calculations for composite samples to complex recipes with multiple ingredients, including cooking.

The food component values for each recipe with a single ingredient (e.g. N1021 - Sausage, beef, pan fried, with no added fat, composite) and recipe with multiple ingredients (e.g. R5424 - Pudding, rice, cocoa & coconut, cooked) are calculated by applying YFs, (Water and Fat) and NRFs (US Department of Agriculture 2007) to the composition of the untreated food(s). Approximately 8% (211 Food Records) of Food Records in the FOODfiles 2018 are derived by recipe calculation: complex recipes (53 Food Records) are only found in the food chapter Recipes (R) and simple recipes (158 Food Records) are listed in associated food chapters (Table 2).

There are 218 Food Records generated by aggregation of similar or distinct Food Records within FIMS, e.g. equal weight portion of similar foods from different brands and/or seasons (e.g. F1028 - Milk, cow, standard 3.3% fat, fluid, composite) or combinations of different parts of food items (e.g. M410 - Pork, shoulder roast, separable lean 74%, hard fat 8%, soft fat 7%, intermuscular fat 6% & skin 5%, raw). They can be found in multiple food chapters (Table 2). YFs and NRFs are considered as 100% in the calculation system.

All ingredient weights are based on edible portion.

2.14.1 Yield Factors (YFs)

Water YF and Fat YF are estimated using the analytical values of water and fat respectively from the raw ingredients and cooked dish or imputed from similar cooked dishes (U.S. Department of Agriculture 2008). The True Weight Yield Factor (WYF) is determined by analysis or borrowed from the published sources (Food Standards Agency 2002a).

Note: The WYF factor mainly represents changes in weight due to water loss/gain and/ or net gain/losses fat during food preparation and cooking. If there is no fat loss/gain during the cooking process, the WYF value is nearly equal to the Water YF. Therefore the WYF value from similar cooked dishes is used as a Water YF value if this is unavailable. The list of the YFs used in the FOODfiles 2018 V 01 can be found in the YF.FT file and are further explained in Section 3.9 below.

The equations for calculating True WYF, Water YF and Fat YF (34–38) are:

Equation 34:

$$\text{True WYF (\%)} = \frac{\text{Weight of the cooked dish (g)}}{\text{Weight of the raw ingredients (g)}} \times 100$$

The True WYF is defined as the measure of the percentage of the weight of the cooked dish in relation to the weight of the raw ingredient/s. The Water YF, Fat YF and NRF equations require the calculation for the True WYF.

Equation 35:

The Water YF is defined as the measure of the proportion of the water in the cooked dish in relation to the water originally present in the raw ingredients.

$$\text{Water YF (\%)} = \frac{\text{Water content in cooked dish (g/100 g)}}{\text{Water contents in raw ingredients (g/100 g)}} \times \text{True WYF (\%)}$$

Equation 36:

The Fat YF is defined as the measure of the proportion of the fat in the cooked dish in relation to the fat originally present in the raw ingredients.

$$\text{Fat YF (\%)} = \frac{\text{Fat content in cooked dish (g/100 g)}}{\text{Fat contents in raw ingredients (g/100 g)}} \times \text{True WYF (\%)}$$

Examples showing calculation of the above YFs can be found in Appendix 7.

2.14.2 Nutrient Retention Factor (NRF)

Nutrient retention is defined as the measure of the proportion of the nutrient remaining in the cooked dish in relation to the nutrient originally present in the raw ingredients.

Equation for calculation of NRF (%):

Equation 37:

$$NRF (\%) = \frac{\text{Nutrient content in cooked dish (g/100 g)}}{\text{Nutrient content in raw ingredients (g/100 g)}} \times \text{True WYF}$$

A set of NRFs used for calculation of nutrients in the cooked dish arising from each raw ingredient can be found in the NRF.FT. They are derived from two sources: U.S. Department of Agriculture (2007) and Food Standards Australia New Zealand (2008). Refer to the INGREDIENT.FT files for Nutrient Retention Factor IDs that are related to the ingredient in the recipe food (cooked dish).

Equation for estimating nutrient content of a recipe (cooked dish):

Equation 38:

$$\begin{aligned} &\text{Nutrient content of cooked dish (per 100 g)} \\ &= (\text{Nutrient content of cooked dish (sum of nutrient content of each raw ingredient} \times \text{NRF)})(g)) \\ &/(\text{Weight of cooked dish (g)}) \times 100 \end{aligned}$$

2.14.3 Recipe calculation system 'Mixed method'

In this method NRF is applied at an ingredient level and the YFs applied at the recipe level (FAO/INFOODS 2012c).

Examples showing calculation of the selected nutrients by the 'Mixed method' can be found in Appendix 8.

3 DOWNLOAD FILES

FOODfiles 2018 V 01 contains four principal data files and four supporting files. Three update files which include foods that have been archived, replaced and new; and components values updated since the release of the FOODfiles 2016 Version 01 are also available.

The eight principal and supporting files listed below are available in two formats: **DSV** (tilde delimited UTF8 encoded) and Microsoft® Office **Excel files**. These files occupy approximately 18 MB and 23 MB, respectively. The Excel files are in **.xlsx** format.

Principal files

- DATA.AP (Standard and unabridged data files) - contains data in a table format)
- DATA.FT (Standard and unabridged data files) - contains data in a list format along with source codes)
- CSM.FT - contains common standard measures along with density
- INGREDIENT.FT contains data on the individual entries used to create composite samples and where relevant the Nutrient Retention Factor ID

Supporting files

- NAME.FT - provides full food details including more detailed description and sampling details
- CODE.FT - provides a list of the components included in the standard and unabridged dataset, their identifier and unit
- NRF.FT - contains the nutrient retention factors used where food have been prepared (e.g. boiled)
- YF.FT - gives the yield factors for foods that have been cooked

Update files:

These three files provide the details of changes from FOODfiles 2016 to 2018.

- New Food Records replacing the old Food Records in latest version of FOODfiles.xlsx
- Food Records archived from latest version of FOODfiles.xlsx
- Data added to or updated in the Food Records in the latest version of FOODfiles.xlsx

Instructions

All files can be found at www.foodcomposition.co.nz and are provided as a Microsoft Windows® installer file (.msi). Use "This PC or "Windows Explorer" to browse to the folder (where you saved the installation program when downloading it) and double click on the installer FOODfilesSetup (.msi) to begin the installation. (NOTE: you may not be able to see the .msi extension, depending on your computer settings). If you do not specify otherwise during the installation, the New Zealand FOODfiles will be installed in the folder **\New Zealand FOODfiles 2018** on your C: disk. If you wish to remove the New Zealand FOODfiles from your computer, you can do so through the Control Panel | All Control Panel | Programs and Features.

3.1 ASCII text files

ASCII text files are delimited and text fields are separated by tildes (~). The tilde characters can be replaced with other field separators. To open these files in Microsoft Excel, go to the "Open file" of the programme and find the folder. Make sure you select "Files of Type". Choose the appropriate file and open it. The "Text Import Wizard" will be activated. In the first screen choose "Delimited" and in the "File Origin" drop-down box, ensure that the entry "Unicode (UTF8)" is selected and press "Next". In the second screen choose "Other" and type "~" in the text box and select "[none]" for "Text Qualifier". Press "Finish" button.

Note that all text fields are separated by a tilde ("~") character. When a field is null or blank then two tilde characters will be adjacent to each other. The first line of all text files contains "© Copyright The New Zealand Institute for Plant and Food Research Limited and the Ministry of Health (New Zealand), 2019 All rights reserved" The field structure starts from the second line of all the text files.

3.2 NAME.FT file

The NAME.FT file contains the FoodID, Food Name, Short Food Name, and descriptions of the foods in New Zealand FOODfiles 2018. The Food Name includes the multi-faceted descriptors according to the INFOODS Food Description System (Truswell et al. 1991). The faceted descriptions are also reported in separate fields. The Short Food Name is used in The Concise New Zealand Food Composition Tables, 13th Edition 2018. Each Short Food Name is unique, represents more common words for the public, and does not typically follow the multi-faceted descriptors from INFOODS.

Each food is uniquely described using the following naming structure in the second line of the text file and in tabular form (Table 9):

Format:

FoodID~Food Name~Short Food Name~AlternativeNames~Food Description~Food Item Generic Name~Food Item Kind~Food Item Part~Process/State~Brand/Strain~Grade~Maturity~Genus~Species~Variety~Sampling Details~Component Message

Example:

X1054~Kumara, root vegetable, root tuber, flesh, raw, gold, Toka Toka~Kumara, Gold, flesh, raw~||USA yam||Sweet potato gold~Commercially harvested fresh sweet potato, medium sweet taste with golden skin.~KUMARA~ROOT VEGETABLE~Root tuber, flesh~Raw~Gold, Toka Toka~~Mature~Ipomoea~batatas~cv.Toko Toko~Samples purchased from various retailers in Auckland, Rotorua, Hawkes Bay, Palmerston North & Christchurch. Analysed January-June 2012.~Fatty acid profile imputed from composite sample of Red, Gold & Orange Kumara.~

Table 9. Example for naming structure of NAME.FT in tabular form

FoodID	X1054
Food Name	Kumara, root vegetable, root tuber, flesh, raw, Gold, Toka Toka
Short Food Name	Kumara, Gold, flesh, raw
Alternative Name	Sweet potato USA yam Sweet potato gold
Food Description	Commercially harvested fresh sweet potato, medium sweet taste with golden skin.
Food Item Generic Name	KUMARA
Food Item Kind	ROOT VEGETABLE
Food Item Part	Root tuber, flesh
Process/State	Raw
Brand/Strain	gold, Toka Toka
Grade	
Maturity	Mature
Genus	<i>Ipomoea</i>
Species	<i>batatas</i>
Variety	cv. Toka Toka
Sampling details	Samples purchased from various retailers in Auckland, Rotorua, Hawkes Bay, Palmerston North & Christchurch. Analysed January-June 2012.
Component Message	Fatty acid profile imputed from composite sample of Red, Gold & Orange Kumara.

The details of the fields included in the NAME.FT file can be found in Table 10.

Table 10. Field definitions of the NAMES.FT file.

Field	Name	Field type alphanumeric	Notes (Required (R) or Optional (O))
1	FoodID	Identifier	(R) The unique code; first character alphabetic, representing food group (Table 2), remaining characters sequential numbers
2	Food Name	Text	(R) Includes the multi-faceted descriptors according to the INFOODS Food Description System (Truswell et al. 1991). Use this field as the 'name' of the food in printed tables and computer packages. Each Food Name is unique
3	Short Food Name	Text	(R) The Short Food Name has fewer words compared to 'Food Name'. The Short Food Name is used in The Concise New Zealand Food Composition Tables. Each Short Food Name is unique, represents more common words for the public, and does not typically follow the multi-faceted descriptors from INFOODS
4	Alternative Name	Text	(O) Other common names to be used for cross-referencing indices
5	Food Description	Text	(O) Detailed information of the food, including ingredients, fortification, appearance, production, preparation and whether the Food Records generated by borrowing from other databases or recipe calculated
6	Food Item Generic Name	Text	(R) Food descriptor facet, most general of food descriptors

Field	Name	Field type alphanumeric	Notes (Required (R) or Optional (O))
7	Food Item Kind	Text	(O) More specific descriptor
8	Food Item Part	Text	(O) Portion of the food represented by the data
9	Process/State	Text	(O) State, condition, treatment, cooking etc.
10	Brand/Strain	Text	(O) More specific descriptor may include brand and trade names where a single product is represented by the Food Record. If the Food Records represented with multiple products such as brands, flavours etc., term 'Composite' is used or derived from recipe calculation 'Recipe' is used
11	Grade	Text	(O) Usually defined by food standards
12	Maturity	Text	(O) Age or stage of development
13	Genus	Text	(O) Latin, first part of scientific name
14	Species	Text	(O) Latin, second part of scientific name
15	Variety	Text	(O) Cultivar name or sub species name
16	Sampling details	Text	(O) Detailed information on samples such as number of samples, product names include brands and flavours, place of purchase, date of purchase and analysis. If the Food Records borrowed from other databases the source and source Food Records information are included
17	Component message	Text	(O) Includes the additional information on Food Composition Data source if not provided in the Data.FT files

3.3 CODE.FT file

The CODE.FT file contains the nutritional component name with its Component Identifier being used in the data files to define the component. Each line is one record and holds information on one component. The units of measure are 'per 100 grams edible portion', unless otherwise specified.

Format:

Component Identifier~Component~Unit
 Example: RETOL~Retinol~µg/100 g

Where:

RETOL is the 'Component Identifier'
 Retinol is the component name of the 'Component Identifier'
 Units of measure is µg per 100 grams edible portion

Appendix 1 contains a full list of 'Component Identifier' and food component names. The details of the fields included in the CODE.FT file can be found in Table 11.

Table 11. Field definitions of the CODES.FT file.

Field	Name	Field type	Notes
1	Component Identifier	Identifier	(Klensin et al. 1989; FAO/INFOODS 2012c)
2	Component	Text	The component name
3	Unit	Text	The units of measure per 100 g edible portion, or as specified

3.4 DATA.FT file

The DATA.FT file contains the data for each component for all foods. Each component forms a record for each food. The format as defined on the second line of the file contains the header:

FoodID~Component Identifier~Value~Source

Each record in the file is composed of a FoodID followed by a field for each of the Component Identifier, the component, value and source codes. Example:

C1022~WATER~86.05~z

Where:

C1022 is the FoodID for Smoothie, berry, ready to drink, composite, fortified vitamin C, related in the file NAME.FT

WATER is the 'Component Identifier' for WATER, related in the file CODES.FT file

86.05 is the value of water in this food

z is the SOURCE code indicating that the information is analytical data from New Zealand laboratories (refer to Appendix 2 for Source Codes)

The details of the fields included in the DATA.FT file can be found in Table 12.

Table 12. Field definitions of DATA.FT.

Field	Name	Field type	Notes
1	FoodID	Identifier	NAME.FT
2	Component Identifier	Identifier	CODE.FT
2	Value	Numeric	Value
3	Source	Identifier	The source of data (Appendix 2)

3.5 DATA.AP file

The DATA.AP file contains some of the information already present in DATA.FT but in an alternative format. It only features the values of the components in each food within rows and columns like a spreadsheet. The second line of the file contains the name of each of the components, and the third line contains the units of measure. The subsequent lines contain a field for each component. These fields are filled by values:

Second line:

FoodID~Chapter~Food Name~Alcohol~Alpha-carotene~Alpha-tocopherol~Ash~Available carbohydrate by difference~Available carbohydrate, FSANZ~Available carbohydrates by weight~Available carbohydrates in monosaccharide equivalent~Beta-carotene~Beta-carotene equivalents~Caffeine~Calcium~Carbohydrate by difference, FSANZ~Cholesterol~Copper~etc.

Third line:

~Chapter~Food
Name~g/100g~µg/100g~mg/100g~g/100g~g/100g~g/100g~g/100g~µg/100g~µg/100g~mg/100g~mg/100g~g/100g~mg/100g~mg/100g~etc

Subsequent lines, an example (27th line):

A1045~A~Bread, mixed grain & seed, sliced, prepacked, Mixed Grain & Toasted Sesame Toast,
Vogel's~0.00~0.00~0.31~1.69~31.33~30.05~30.05~33.04~0.00~0.00~0.00~23.00~31.33~0.00~0.18~etc.

Table 13 describes the example of the DATA.FT in tabular form.

Table 13. Example for DATA.FT of A1045 in tabular form.

Field	Second line	Third line	Fifth line (example)
1	FoodID	Null or blank	A1045
2	Chapter	Chapter	A
3	Food Name	Food name	Bread, mixed grain & seed, sliced, prepacked, Mixed Grain & Toasted Sesame Toast, Vogel's
4	Alcohol	g/100 g	0.00
5	Alpha-carotene	µg/100 g	0.00
6	Alpha-tocopherol	mg/100 g	0.31
7	Ash	g/100 g	1.69
8	Available carbohydrate by difference	g/100 g	31.33
9	Available carbohydrate, FSANZ	g/100 g	30.05
10	Available carbohydrates by weight	g/100 g	30.05
11	Available carbohydrates in monosaccharide equivalent	g/100 g	33.04
12	Beta-carotene	µg/100 g	0.00
13	Beta-carotene equivalents	µg/100 g	0.00
14	Caffeine	mg/100 g	0.00
15	Calcium	mg/100 g	23.00
16	Carbohydrate by difference, FSANZ	g/100 g	31.33

Field	Second line	Third line	Fifth line (example)
17	Cholesterol	mg/100 g	0.00
18	Copper	mg/100 g	0.18
	etc.		

3.6 CSM.FT file

The CSM.FT file contains the data for the common standard measures (CSM) and density for most foods. The CSM is either expressed as New Zealand metric standards (Appendix 10) or as the amount commonly purchased or eaten. All these measure descriptions are listed under the CSM column, noting that there may be more than one CSM for any particular food. All the measures (g) were made on an edible portion, so no adjustments are necessary to account for inedible portion. For the alcoholic beverages in New Zealand, one standard drink containing 10 grams of pure alcohol is used (Health Promotion Agency. April 2016).

Density is measured as specific gravity, mass density, bulk density and/or use mass and volume by water or rapeseed displacement method. Mass density of a food is defined as its mass per unit volume. Specific gravity is for liquid foods (e.g. milk, fruit juices). Mass density is typically for semi solids and solids with regular shapes (e.g. bread, biscuits). Bulk density is the typical measurement for powders, grains, or other particulate solids that contain air. The bulk density total volume includes particle volume, inter-particle void volume and internal pore volume (e.g. Almonds ground and slivered). Water displacement method is used for foods with irregular shapes (e.g. eggplant, meat cuts) in order to establish the volume that relates to an item of known mass/weight to allow density to be calculated. Similarly, baked product can be measured using rapeseed instead of water.

The format, as defined on the second line of the CSM.FT file, is:

FoodID~Food Name~CSM~Measure (g)~Density
(g/cm³)>

For example:

A1045~Bread, mixed grain & seed, sliced, prepacked, Mixed Grain & Toasted Sesame Toast, Vogel's~1 slice toast (9.5 x 10.3 x 1.2 cm)~42.50~0.37

In the example above:

A1045 is the FoodID for a Food Name, Bread, mixed grain & seed, sliced, prepacked, Mixed Grain & Toasted Sesame Toast, Vogel's

1 slice toast (9.5 x 10.3 x 1.2 cm) of A1045 weights 42.50 grams and has a density of 0.37 g/cm³

NOTE: The weight specified in the Measure (g) field can be used as a percent scaling factor to convert the component mean values in the DATA.FT and DATA.AP data files to component values per CSM (New Zealand Food Composition Database 2019) as all values in DATA.FT and DATA.AP are expressed per 100 g.

The details of the fields included in the CSM.FT file can be found in Table 14.

Table 14. Field definitions of the file CSM.FT.

Field	Name	Field type	Notes
1	Food ID	Identifier	NAME.FT file
2	Food Name	Text	NAME.FT file
3	CSM	Text	The description of the Common Standard Measure (CSM) or amount commonly purchased or eaten
4	Measure (g)	Numeric	The weight of the CSM amount commonly purchased or eaten in grams
5	Density	Numeric	g/cm ³ , Food Records may have multiple density based on the physical state e.g. 1 cup (250 mL) almond: ground slivered and whole

3.7 INGREDIENT.FT file

The INGREDIENTS.FT file contains the FoodID of a recipe, FoodIDs of the ingredients, its Food Names, Weight Fractions used in that particular recipe, Nutrition Retention Factor ID and USDA Retention Factor description. The Nutrition Retention Factor ID is specific to the cooking method (USDA Retention Factor description).

The naming structure for each record in this file is:

RecipeID~IngredientID~Ingredient Name~Weight Fraction(%)~Nutrient Retention Factor ID~USDA Retention factor description

For example the recipe 'Pie, lemon meringue, baked' (R5442) contains the following:

R5442~W20~Sugar, caster~22.7~~
R5442~F1046~Butter, salted, composite~1.5~~
R5442~L184~Lemon, peel, raw~1.3~P162~FRUITS,CITRUS,CKD
R5442~E9~Flour, corn~3.5~P152~FLOUR/MEAL,BAKED
R5442~R5402~Pastry, short, butter, raw~21.0~P152~FLOUR/MEAL,BAKED
R5442~G1008~Egg, chicken, white & yolk, raw~16.2~P128~EGGS,BAKED
R5442~L91~Juice, lemon, raw~14.0~~
R5442~C40~Water, tap~19.7~~

The tabulated form of the above example R5442 can be found in Table 15.

Table 15. Example for INGREDIENT.FT of R5442 in tabular form.

Recipe FoodID	Ingredient FoodID	Ingredient name	Weight fraction (g) ^a	Nutrient retention factor ID	USDA retention factor description ^b
R5442	W20	Sugar, caster	22.7		
R5442	F1046	Butter, salted, composite	1.5		
R5442	L184	Lemon, peel, raw	1.3	P228	FRUITS,CITRUS,CKD
R5442	E9	Flour, corn	3.5	P152	FLOUR/MEAL,BAKED
R5442	R5402	Pastry, short, butter, raw	21.0	P152	FLOUR/MEAL,BAKED
R5442	G1008	Egg, chicken, white & yolk, raw	16.2	P128	EGGS,BAKED
R5442	L91	Juice, lemon, raw	14.0		
R5442	C40	Water, tap	19.7		

^a Weight fraction is the percentage of each ingredient used in the recipe^b US Department of Agriculture (2007).

The details of the fields included in the INGREDIENT.FT file can be found in Table 16.

Table 16. Field definitions of the file INGREDIENT.FT.

Field	Name	Field type	Notes
1	RecipeID	Identifier	NAME.FT file
2	Ingredient ID	Identifier	NAME.FT file
3	Ingredient Name	Text	NAME.FT file
4	Weight Fraction	Numeric	The weight fraction of the food ingredient in the recipe
5	Nutrient Retention Factor ID	Numeric	First character alphabetic (P), remaining characters three digit sequential numbers. Refer NRF.FT file
6	USDA Retention factor description	Identifier	U.S. Department of Agriculture (2007)

3.8 NRF.FT

The NRF.FT contains the Nutrient Retention Factor for selected nutrients and ID. The naming structure in this file is:

Nutrient Retention Factor ID~Component Identifier~Nutrient Retention Factor
Example: P218~FOL~0.7

Where:

P218 is the 'Nutrient Retention Factor ID' for the USDA Retention factor description of "PASTA,BOILED,DRAINED", related to the INGREDIENT.FT

FOL is the 'Component Identifier' for 'Folate, total', related in the file CODES.FT file'

0.65 is 'Nutrient Retention Factor'

The details of the fields included in the NRF.FT file can be found in Table 17.

Table 17. Field definitions of the file NRF.FT.

Field	Name	Field type	Notes
1	Nutrient Retention Factor ID	Numeric	First character alphabetic (P), remaining characters three digit sequential numbers. Related to the INGREDIENT.FT
2	Component Identifier	Identifier	CODE.FT file
3	Nutrient Retention Factor	Numeric	Ratio of the nutrient retention between cooked and raw ingredients (g/g) U.S. Department of Agriculture (2007); (Food Standards Australia New Zealand 2008)

3.9 YF.FT

The YF.FT contains the Water YF, Fat YF and Weight YF for recipe Food Records.

The naming structure in this file is:

RecipeID~Recipe Process~Water Yield Factor~FAT Yield Factor~Weight Yield Factor~Source

The details of the fields included in the YF.FT file can be found in Table 18.

Table 18. Field definitions of the file YF.FT.

Field	Name	Field type	Notes
1	RecipeID	Identifier	INGREDIENT.FT file
2	Recipe Process	Text	Process for making recipe
3	Water Yield Factor	Numeric	Measure of the proportion of the water in the cooked dish in relation to the water originally present in the raw ingredients (Equation 35)
4	Fat Yield Factor	Numeric	Measure of the proportion of the fat in the cooked dish in relation to the fat originally present in the raw ingredients (Equation 36)
5	Weight Yield Factor	Numeric	Measure of the percentage of the weight of the cooked dish in relation to the weight of the raw ingredient/s (Equation 34)
6	Source	Identifier	The source of data (Appendix 2) for Weight Yield Factor

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APPENDICES

Appendix 1. List of components with supporting details

This table provides details of the components included in the database including which publications they are included in.

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
ACEAC_G	Acetic acid (grams)	ACEAC_G	g/100 g	A				80	3
AL	Aluminium	AL	µg/100 g	A			✓	544	20
ALA	Alanine	ALA	mg/100 g	A			✓	595	22
ALA_G	Alanine (grams)	ALA_G	g/100 g	A			✓	594	22
ALAN	Alanine (/g N)	ALAN	mg/g N	C			✓	594	22
ALC	Alcohol	ALC	g/100 g	A	✓	✓	✓	2767	100
ARG	Arginine	ARG	mg/100 g	A			✓	597	22
ARG_G	Arginine (grams)	ARG_G	g/100 g	A			✓	596	22
ARGN	Arginine (/g N)	ARGN	mg/g N	C			✓	596	22
AS	Arsenic	AS	µg/100 g	A			✓	558	20
ASH	Ash	ASH	g/100 g	A		✓	✓	2767	100
ASN	Asparagine	ASN	mg/100 g	A			✓	178	6
ASN_G	Asparagine (grams)	ASN_G	g/100 g	A			✓	178	6
ASNN	Asparagine (/g N)	ASNN	mg/g N	C			✓	178	6
ASP	Aspartic acid	ASP	mg/100 g	A			✓	417	15
ASP_G	Aspartic acid	ASP_G	g/100 g	A			✓	416	15
ASPN	Aspartic acid	ASPN	mg/g N	C			✓	416	15
B	Boron	B	µg/100 g	A			✓	574	21
BIOT	Biotin	BIOT	µg/100 g	A			✓	1297	47
CA	Calcium	CA	mg/100 g	A	✓	✓	✓	2767	100
CAFFN	Caffeine	CAFFN	mg/100 g	A	✓	✓	✓	2767	100
CARTA	Alpha-carotene	CARTA	µg/100 g	A		✓	✓	2224	80
CARTB	Beta-carotene	CARTB	µg/100 g	A		✓	✓	2393	86
CARTBEQ	Beta-carotene equivalents	CARTBEQ	µg/100 g	C	✓	✓	✓	2767	100
CD	Cadmium	CD	µg/100 g	A			✓	682	25
CHOAVL	Available carbohydrates by weight	CHOAVL	g/100 g	C		✓	✓	2767	100
CHOAVL_FSANZ	Available carbohydrate		g/100 g	C	✓	✓	✓	2767	100

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
CHOAVLDF	Available carbohydrate by difference	CHOAVLDF	g/100 g	C		✓	✓	2767	100
CHOAVLDF_FSANZ	Carbohydrate by difference		g/100 g	C		✓	✓	2767	100
CHOAVLM	Available carbohydrates in monosaccharide equivalent	CHOAVLM	g/100 g	C		✓	✓	2767	100
CHOCAL	Cholecalciferol (Vitamin D3)	CHOCAL	µg/100 g	A			✓	2767	100
CHOCALOH	25-hydroxyvitamin D3	CHOCALOH	µg/100 g	A			✓	190	7
CHOCDF	Total carbohydrate by difference	CHOCDF	g/100 g	C		✓	✓	2767	100
CHOCSM	Total carbohydrates by summation	CHOCSM	g/100 g	C		✓	✓	2767	100
CHOLE	Cholesterol	CHOLE	mg/100 g	A	✓	✓	✓	2767	100
CITAC_G	Citric acid (grams)	CITAC_G	g/100 g	A			✓	53	2
CLD	Chloride	CLD	mg/100 g	A			✓	1133	41
CO	Cobalt	CO	µg/100 g	A			✓	498	18
CR	Chromium	CR	µg/100 g	A			✓	629	23
CS	Caesium	CS	µg/100 g	A			✓	401	14
CU	Copper	CU	mg/100 g	A		✓	✓	2767	100
CYS	Cystine	CYS	mg/100 g	A			✓	580	21
CYS_G	Cystine (grams)	CYS_G	g/100 g	A			✓	579	21
CYSN	Cystine (/g N)	CYSN	mg/g N	C			✓	579	21
DISAC	Disaccharides, total	DISAC	g/100 g	C			✓	2767	100
DISACM	Disaccharides, total (monosaccharide equivalents)	DISACM	g/100 g	C			✓	2767	100
DM	Dry matter	DM	g/100 g	C		✓	✓	2767	100
EDIBLE	Edible portion	EDIBLE	%	A		✓	✓	2767	100
ENERC	Energy, total metabolisable (kJ)	ENERC	kJ/100 g	C	✓	✓	✓	2767	100
ENERC1	Energy, total metabolisable (kJ, including dietary fibre)		kJ/100 g	C		✓	✓	2767	100
ENERC_KCAL	Energy, total metabolisable (kcal)	ENERC_KCAL	kcal/100 g	C		✓	✓	2767	100
ENERC1_KCAL	Energy, total metabolisable (kcal, including dietary fibre)		kcal/100 g	C		✓	✓	2767	100
ENERC_FSANZ1	Energy, total metabolisable, carbohydrate by difference, FSANZ (kJ)		kJ/100 g	C		✓	✓	2767	100

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
ENERC_FSANZ2	Energy, total metabolisable, available carbohydrate, FSANZ (kJ)		kJ/100 g	C	✓	✓	✓	2767	100
ENERC_FSANZ1_KCAL	Energy, total metabolisable, carbohydrate by difference, FSANZ (kcal)		kcal/100 g	C		✓	✓	2767	100
ENERC_FSANZ2_KCAL	Energy, total metabolisable, available carbohydrate, FSANZ (kcal)		kcal/100 g	C		✓	✓	2767	100
ERGCAL	Ergocalciferol (Vitamin D2)	ERGCAL	µg/100 g	A			✓	2767	100
F4D0	Fatty acid 4:0	F4D0	g/100 g	C			✓	631	23
F4D0F	Fatty acid 4:0 (/100 g TFA)	F4D0F	g/100 g TFA	A			✓	597	22
F6D0	Fatty acid 6:0	F6D0	g/100 g	C			✓	1025	37
F6D0F	Fatty acid 6:0 (/100 g TFA)	F6D0F	g/100 g TFA	A			✓	1025	37
F8D0	Fatty acid 8:0	F8D0	g/100 g	C			✓	1235	45
F8D0F	Fatty acid 8:0 (/100 g TFA)	F8D0F	g/100 g TFA	A			✓	1201	43
F10D0	Fatty acid 10:0	F10D0	g/100 g	C			✓	1362	49
F10D0F	Fatty acid 10:0 (/100 g TFA)	F10D0F	g/100 g TFA	A			✓	1289	47
F12D0	Fatty acid 12:0	F12D0	g/100 g	C			✓	1433	52
F12D0F	Fatty acid 12:0 (/100 g TFA)	F12D0F	g/100 g TFA	A			✓	1369	50
F13D0	Fatty acid 13:0	F13D0	g/100 g	C			✓	1050	38
F13D0F	Fatty acid 13:0 (/100 g TFA)	F13D0F	g/100 g TFA	A			✓	1009	36
F14D0	Fatty acid 14:0	F14D0	g/100 g	C			✓	1846	67
F14D0F	Fatty acid 14:0 (/100 g TFA)	F14D0F	g/100 g TFA	A			✓	1766	64
F15D0	Fatty acid 15:0	F15D0	g/100 g	C			✓	955	35
F15D0F	Fatty acid 15:0 (/100 g TFA)	F15D0F	g/100 g TFA	A			✓	891	32
F16D0	Fatty acid 16:0	F16D0	g/100 g	C			✓	1938	70
F16D0F	Fatty acid 16:0 (/100 g TFA)	F16D0F	g/100 g TFA	A			✓	1851	67
F17D0	Fatty acid 17:0	F17D0	g/100 g	C			✓	1624	59
F17D0F	Fatty acid 17:0 (/100 g TFA)	F17D0F	g/100 g TFA	A			✓	1560	56
F18D0	Fatty acid 18:0	F18D0	g/100 g	C			✓	1936	70

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
F18D0F	Fatty acid 18:0 (/100 g TFA)	F18D0F	g/100 g TFA	A			✓	1849	67
F19D0	Fatty acid 19:0	F19D0	g/100 g	C			✓	41	2
F19D0F	Fatty acid 19:0 (/100 g TFA)	F19D0F	g/100 g TFA	A			✓	38	1
F20D0	Fatty acid 20:0	F20D0	g/100 g	C			✓	1499	54
F20D0F	Fatty acid 20:0 (/100 g TFA)	F20D0F	g/100 g TFA	A			✓	1433	52
F21D0	Fatty acid 21:0	F21D0	g/100 g	C			✓	615	22
F21D0F	Fatty acid 21:0 (/100 g TFA)	F21D0F	g/100 g TFA	A			✓	615	22
F22D0	Fatty acid 22:0	F22D0	g/100 g	C			✓	1266	46
F22D0F	Fatty acid 22:0 (/100 g TFA)	F22D0F	g/100 g TFA	A			✓	1230	45
F23D0	Fatty acid 23:0	F23D0	g/100 g	C			✓	590	21
F23D0F	Fatty acid 23:0 (/100 g TFA)	F23D0F	g/100 g TFA	A			✓	590	21
F24D0	Fatty acid 24:0	F24D0	g/100 g	C			✓	1188	43
F24D0F	Fatty acid 24:0 (/100 g TFA)	F24D0F	g/100 g TFA	A			✓	1129	41
FASAT	Fatty acids, total saturated	FASAT	g/100 g	C	✓	✓	✓	2767	100
FASATF	Fatty acids, total saturated (/100 g TFA)	FASATF	g/100 g TFA	C			✓	1955	71
F10D1	Fatty acid 10:1	F10D1	g/100 g	C			✓	515	19
F10D1F	Fatty acid 10:1 (/100 g TFA)	F10D1F	g/100 g TFA	A			✓	485	18
F12D1	Fatty acid 12:1	F12D1	g/100 g	C			✓	255	9
F12D1F	Fatty acid 12:1 (/100 g TFA)	F12D1F	g/100 g TFA	A			✓	183	7
F14D1N5	Fatty acid 14:1 omega-5	F14D1N5	g/100 g	C			✓	666	24
F14D1N5F	Fatty acid 14:1 omega-5 (/100 g TFA)	F14D1FN5F	g/100 g TFA	A			✓	663	24
F14D1	Fatty acid 14:1	F14D1	g/100 g	C			✓	1420	51
F14D1F	Fatty acid 14:1 (/100 g TFA)	F14D1F	g/100 g TFA	C			✓	1371	50
F15D1	Fatty acid 15:1	F15D1	g/100 g	C			✓	1197	43
F15D1F	Fatty acid 15:1 (/100 g TFA)	F15D1F	g/100 g TFA				✓	1141	41
F16D1C	Fatty acid <i>cis</i> 16:1	F16D1C	g/100 g	C			✓	1244	45
F16D1CF	Fatty acid <i>cis</i> 16:1 (/100 g TFA)	F16D1CF	g/100 g TFA	A			✓	1202	43
F16D1T	Fatty acid <i>trans</i> 16:1	F16D1T	g/100 g	C			✓	714	26

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
F16D1TF	Fatty acid <i>trans</i> 16:1 (/100 g TFA)	F16D1TF	g/100 g TFA	A			✓	692	25
F16D1	Fatty acid 16:1	F16D1	g/100 g	C			✓	1825	66
F16D1F	Fatty acid 16:1 (/100 g TFA)	F16D1F	g/100 g TFA	C			✓	1723	62
F17D1	Fatty acid 17:1	F17D1	g/100 g	C			✓	1589	57
F17D1F	Fatty acid 17:1 (/100 g TFA)	F17D1F	g/100 g TFA	A			✓	1524	55
F18D1CN7	Fatty acid <i>cis</i> 18:1 omega-7	F18D1CN7	g/100 g	C			✓	990	36
F18D1CN7F	Fatty acid <i>cis</i> 18:1 omega-7 (/100 g TFA)	F18D1CN7F	g/100 g TFA	A			✓	959	35
F18D1TN7	Fatty acid <i>trans</i> 18:1 omega-7	F18D1CN7	g/100 g	C			✓	615	22
F18D1TN7F	Fatty acid <i>trans</i> 18:1 omega-7 (/100 g TFA)	F18D1CN7F	g/100 g TFA	A			✓	615	22
F18D1N7	Fatty acid 18:1 omega-7	F18D1N7	g/100 g	C			✓	1187	43
F18D1N7F	Fatty acid 18:1 omega-7 (/100 g TFA)	F18D1N7F	g/100 g TFA	C			✓	1160	42
F18D1CN9	Fatty acid <i>cis</i> 18:1 omega-9	F18D1CN9	g/100 g	C			✓	1210	44
F18D1CN9F	Fatty acid <i>cis</i> 18:1 omega-9 (/100 g TFA)	F18D1CN9F	g/100 g TFA	A			✓	1180	43
F18D1TN9	Fatty acid <i>trans</i> 18:1 omega-9	F18D1TN9	g/100 g	C			✓	1007	36
F18D1TN9F	Fatty acid <i>trans</i> 18:1 omega-9 (/100 g TFA)	F18D1TN9F	g/100 g TFA	A			✓	992	36
F18D1N9	Fatty acid 18:1 omega-9	F18D1N9	g/100 g	C			✓	1241	45
F18D1N9F	Fatty acid 18:1 omega-9 (/100 g TFA)	F18D1N9F	g/100 g TFA	C			✓	1210	44
F18D1C	Fatty acid <i>cis</i> 18:1	F18D1C	g/100 g	C			✓	1267	46
F18D1CF	Fatty acid <i>cis</i> 18:1 (/100 g TFA)	F18D1CF	g/100 g TFA	C			✓	1234	44
F18D1T	Fatty acid <i>trans</i> 18:1	F18D1T	g/100 g	C			✓	1088	39
F18D1TF	Fatty acid <i>trans</i> 18:1 (/100 g TFA)	F18D1TF	g/100 g TFA	C			✓	1050	38
F18D1	Fatty acid 18:1	F18D1	g/100 g	C			✓	1982	72
F18D1F	Fatty acid 18:1 (/100 g TFA)	F18D1F	g/100 g TFA	C			✓	1924	70
F20D1N9	Fatty acid 20:1 omega-9	F20D1N9	g/100 g	C			✓	1105	40
F20D1N9F	Fatty acid 20:1 omega-9 (/100 g TFA)	F20D1N9F	g/100 g TFA	A			✓	1080	39
F20D1N11	Fatty acid 20:1 omega-11	F20D1N11	g/100 g	C			✓	333	12
F20D1N11F	Fatty acid 20:1 omega-11 (/100 g TFA)	F20D1N11F	g/100 g TFA	A			✓	325	11

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
F20D1	Fatty acid 20:1	F20D1	g/100 g	C			✓	1605	58
F20D1F	Fatty acid 20:1 (/100 g TFA)	F20D1F	g/100 g TFA	C			✓	1499	54
F22D1N9	Fatty acid 22:1 omega-9	F22D1N9	g/100 g	C			✓	790	28
F22D1N9F	Fatty acid 22:1 omega-9 (/100 g TFA)	F22D1N9F	g/100 g TFA	A			✓	673	24
F22D1N11	Fatty acid 22:1 omega-11	F22D1N11	g/100 g	C			✓	395	14
F22D1N11F	Fatty acid 22:1 omega-11 (/100 g TFA)	F22D1N11F	g/100 g TFA	A			✓	356	44
F22D1	Fatty acid 22:1	F22D1	g/100 g	C			✓	1260	45
F22D1F	Fatty acid 22:1 (/100 g TFA)	F22D1F	g/100 g TFA	C			✓	1220	44
F24D1	Fatty acid 24:1	F24D1	g/100 g	C			✓	1057	38
F24D1F	Fatty acid 24:1 (/100 g TFA)	F24D1F	g/100 g TFA	A			✓	1000	36
FAMST	Fatty acids, total monounsaturated <i>trans</i>	FAMST	g/100 g	C			✓	1153	42
FAMSTF	Fatty acids, total monounsaturated <i>trans</i> (/100 g TFA)	FAMSTF	g/100 g TFA	C			✓	1147	41
FAMS	Fatty acids, total monounsaturated	FAMS	g/100 g	C	✓	✓	✓	2767	100
FAMSF	Fatty acids, total monounsaturated (/100 g TFA)	FAMSF	g/100 g TFA	C			✓	1954	71
F18D2CN6	Fatty acid <i>cis,cis</i> 18:2 omega-6	F18D2CN6	g/100 g	C	✓	✓	✓	1469	53
F18D2CN6F	Fatty acid <i>cis,cis</i> 18:2 omega-6 (/100 g TFA)	F18D2CN6F	g/100 g TFA	A			✓	1416	51
F18D2TN6	Fatty acid <i>trans</i> 18:2 omega-6	F18D2TN6	g/100 g	C			✓	1009	36
F18D2TN6F	Fatty acid <i>trans</i> 18:2 omega-6 (/100 g TFA)	F18D2TN6F	g/100 g TFA	A			✓	930	34
F18D2N6	Fatty acid 18:2 omega-6	F18D2N6	g/100 g	C			✓	1532	55
F18D2N6F	Fatty acid 18:2 omega-6 (/100 g TFA)	F18D2N6F	g/100 g TFA	C			✓	1422	51
F18D2CN9T11	Fatty acid <i>cis, trans</i> 18:2 omega-9, 11		g/100 g	C		✓	✓	917	33
F18D2CN9TN11F	Fatty acid <i>cis, trans</i> 18:2 omega-9, 11 (/100 g TFA)		g/100 g TFA	A			✓	826	30
F18D2C	Fatty acid <i>cis</i> 18:2	F18D2C	g/100 g	C			✓	1464	53
F18D2CF	Fatty acid <i>cis</i> 18:2 (/100 g TFA)	F18D2NCF	g/100 g TFA	C			✓	1416	51
F18D2T	Fatty acid <i>trans</i> 18:2	F18D2T	g/100 g	C			✓	1150	42

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
F18D2TF	Fatty acid <i>trans</i> 18:2 (/100 g TFA)	F18D2TF	g/100 g TFA	C			✓	1113	40
F18D2	Fatty acid 18:2	F18D2	g/100 g	C			✓	1935	70
F18D2F	Fatty acid 18:2 (/100 g TFA)	F18D2F	g/100 g TFA	C			✓	1851	67
F18D3N3	Fatty acid 18:3 omega-3	F18D3N3	g/100 g	C	✓	✓	✓	1390	50
F18D3N3F	Fatty acid 18:3 omega-3 (/100 g TFA)	F18D3N3F	g/100 g TFA	A			✓	1284	46
F18D3TN3	Fatty acid <i>trans</i> 18:3 omega-3	F18D3TN3	g/100 g	C			✓	306	11
F18D3TN3F	Fatty acid <i>trans</i> 18:3 omega-6 (/100 g TFA)	F18D3TN3F	g/100 g TFA	A			✓	306	11
F18D3N6	Fatty acid 18:3 omega-6	F18D3N6	g/100 g	C			✓	1053	38
F18D3N6F	Fatty acid 18:3 omega-6 (/100 g TFA)	F18D3N6F	g/100 g TFA	A			✓	1026	37
F18D3	Fatty acid 18:3	F18D3	g/100 g	C			✓	1768	64
F18D3F	Fatty acid 18:3 (/100 g TFA)	F18D3F	g/100 g TFA	C			✓	1683	61
F18D4N3	Fatty acid 18:4 omega-3	F18D4N3	g/100 g	C			✓	323	12
F18D4N3F	Fatty acid 18:4 omega-3 (/100 g TFA)	F18D4N3F	g/100 g TFA	A			✓	317	11
F18D4	Fatty acid 18:4	F18D4	g/100 g	C			✓	347	12
F18D4F	Fatty acid 18:4 (/100 g TFA)	F18D4F	g/100 g TFA	C			✓	338	12
F20D2N6	Fatty acid 20:2 omega-6	F20D2N6	g/100 g	C			✓	928	34
F20D2N6F	Fatty acid 20:2 omega-6 (/100 g TFA)	F20D2N6F	g/100 g TFA	A			✓	911	33
F20D2	Fatty acid 20:2	F20D2	g/100 g	C			✓	1172	42
F20D2F	Fatty acid 20:2 (/100 g TFA)	F20D2F	g/100 g TFA	C			✓	1115	40
F20D3N3	Fatty acid 20:3 omega-3	F20D3N3	g/100 g	C			✓	886	32
F20D3N3F	Fatty acid 20:3 omega-3 (/100 g TFA)	F20D3N3F	g/100 g TFA	A			✓	879	32
F20D3N6	Fatty acid 20:3 omega-6	F20D3N6	g/100 g	C			✓	1053	38
F20D3N6F	Fatty acid 20:3 omega-6 (/100 g TFA)	F20D3N6F	g/100 g TFA	A			✓	1047	38
F20D3	Fatty acid 20:3	F20D3	g/100 g	C			✓	1322	48
F20D3F	Fatty acid 20:3 (/100 g TFA)	F20D3F	g/100 g TFA	C			✓	1321	48
F20D4N3	Fatty acid 20:4 omega-3	F20D4N3	g/100 g	C			✓	384	14
F20D4N3F	Fatty acid 20:4 omega-3 (/100 g TFA)	F20D4N3F	g/100 g TFA	A			✓	357	13
F20D4N6	Fatty acid 20:4 omega-6	F20D4N6	g/100 g	C			✓	1078	39

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
F20D4N6F	Fatty acid 20:4 omega-6 (/100 g TFA)	F20D4N6F	g/100 g TFA	A			✓	1054	38
F20D4	Fatty acid 20:4	F20D4	g/100 g	C			✓	1405	51
F20D4F	Fatty acid 20:4 (/100 g TFA)	F20D4F	g/100 g TFA	C		✓	✓	1346	49
F20D5N3	Fatty acid 20:5 omega-3	F20D5N3	g/100 g	C		✓	✓	1101	40
F20D5N3F	Fatty acid 20:5 omega-3 (/100 g TFA)	F20D5N3F	g/100 g TFA	A			✓	1070	39
F20D5	Fatty acid 20:5	F20D5	g/100 g	C			✓	1336	48
F20D5F	Fatty acid 20:5 (/100 g TFA)	F20D5F	g/100 g TFA	C			✓	1296	47
F21D5N3	Fatty acid 21:5 omega-3	F21D5N3	g/100 g	C			✓	164	6
F21D5N3F	Fatty acid 21:5 omega-3 (/100 g TFA)	F21D5N3F	g/100 g TFA	A			✓	159	6
F21D5	Fatty acid 21:5	F21D5	g/100 g	C			✓	249	9
F21D5F	Fatty acid 21:5 (/100 g TFA)	F21D5F	g/100 g TFA	C			✓	247	9
F22D2N6	Fatty acid 22:2 omega-6	F22D2N6	g/100 g	C			✓	621	22
F22D2N6F	Fatty acid 22:2 omega-6 (/100 g TFA)	F22D2N6F	g/100 g TFA	A			✓	621	22
F22D2	Fatty acid 22:2	F22D2	g/100 g	C			✓	682	25
F22D2F	Fatty acid 22:2 (/100 g TFA)	F22D2F	g/100 g TFA	C			✓	678	25
F22D4N6	Fatty acid 22:4 omega-6	F22D4N6	g/100 g	C			✓	220	8
F22D4N6F	Fatty acid 22:4 omega-6 (/100 g TFA)	F22D4N6F	g/100 g TFA	A			✓	211	8
F22D4	Fatty acid 22:4	F22D4	g/100 g	C			✓	378	14
F22D4F	Fatty acid 22:4 (/100 g TFA)	F22D4F	g/100 g TFA	C			✓	378	14
F22D5N3	Fatty acid 22:5 omega-3	F22D5N3	g/100 g	C		✓	✓	1088	39
F22D5N3F	Fatty acid 22:5 omega-3 (/100 g TFA)	F22D5N3F	g/100 g TFA	A			✓	1051	38
F22D5N6	Fatty acid 22:5 omega-6	F22D5N6	g/100 g	C			✓	374	14
F22D5N6F	Fatty acid 22:5 omega-6 (/100 g TFA)	F22D5N6F	g/100 g TFA	A			✓	370	13
F22D5	Fatty acid 22:5	F22D5	g/100 g	C			✓	1289	46
F22D5F	Fatty acid 22:5 (/100 g TFA)	F22D5F	g/100 g TFA	C			✓	1288	46
F22D6N3	Fatty acid 22:6 omega-3	F22D6N3	g/100 g	C		✓	✓	1085	39
F22D6N3F	Fatty acid 22:6 omega-3 (/100 g TFA)	F22D6N3F	g/100 g TFA	A			✓	1061	38
F22D6	Fatty acid 22:6	F22D6	g/100 g	C			✓	1413	51
F22D6F	Fatty acid 22:6 (/100 g TFA)	F22D6F	g/100 g TFA	C			✓	1355	49

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
FAPUN3	Fatty acids, total polyunsaturated omega-3	FAPUN3	g/100 g	C		✓	✓	1454	52
FAPUN3F	Fatty acids, total polyunsaturated omega-3 (/100 g TFA)	FAPUN3F	g/100 g TFA	C			✓	1347	49
FAPUN6	Fatty acids, total polyunsaturated omega-6	FAPUN6	g/100 g	C		✓	✓	1553	56
FAPUN6F	Fatty acids, total polyunsaturated omega-6 (/100 g TFA)	FAPUN6F	g/100 g TFA	C			✓	1444	52
FALCPUN3	Fatty acids, total long chain polyunsaturated omega-3		g/100 g	C		✓	✓	1183	43
FALCPUN3F	Fatty acids, total long chain polyunsaturated omega-3 (/100 g TFA)		g/100 g TFA	C			✓	1148	41
FAPUT	Fatty acids, total polyunsaturated <i>trans</i>	FAPUT	g/100 g	C			✓	1141	41
FAPUTF	Fatty acids, total polyunsaturated <i>trans</i> (/100 g TFA)	FAPUTF	g/100 g TFA	C			✓	1140	41
FAPU	Fatty acids, total polyunsaturated	FAPU	g/100 g	C	✓	✓	✓	2767	100
FAPUF	Fatty acids, total polyunsaturated (/100 g TFA)	FAPUF	g/100 g TFA	C			✓	1953	71
FATR	Fatty acids, total <i>trans</i>	FATR	g/100 g	C		✓	✓	1186	43
FATRNF	Fatty acids, total <i>trans</i> (/100 g TFA)	FATRNF	g/100 g TFA	C			✓	1184	43
FACID	Fatty acids, total	FACID	g/100 g	A or C			✓	2767	100
FAT	Fat, total	FAT	g/100 g	A	✓	✓	✓	2767	100
FD	Fluoride	FD	µg/100 g	A			✓	34	1
FE	Iron	FE	mg/100 g	A	✓	✓	✓	2767	100
FIBINS	Fibre, water-insoluble	FIBINS	g/100 g	A		✓	✓	1730	62
FIBSOL	Fibre, water-soluble	FIBSOL	g/100 g	A		✓	✓	1730	62
FIBTG	Fibre, total dietary	FIBTG	g/100 g	A	✓	✓	✓	2767	100
FOL	Folate, total	FOL	µg/100 g	A		✓	✓	2767	100
FOLAC	Folic acid, synthetic folic acid	FOLAC	µg/100 g	A		✓	✓	2767	100
FOLDFE	Dietary folate equivalents	FOLDFE	µg/100 g	C	✓	✓	✓	2767	100
FOLFD	Folate food, naturally occurring food folates	FOLFD	µg/100 g	C		✓	✓	2767	100
FRUS	Fructose	FRUS	g/100 g	A		✓	✓	2767	100

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
GLU	Glutamic acid	GLU	mg/100 g	A			✓	593	21
GLU_G	Glutamic acid (grams)	GLU_G	g/100 g	A			✓	592	21
GLUN	Glutamic acid (/g N)	GLUN	mg/g N	C			✓	592	21
GLUS	Glucose	GLUS	g/100 g	A		✓	✓	2767	100
GLY	Glycine	GLY	mg/100 g	A			✓	499	18
GLY_G	Glycine (grams)	GLY_G	g/100 g	A			✓	498	18
GLYN	Glycine (/g N)	GLYN	mg/g N	C			✓	498	18
GLYC	Glycogen	GLYC	g/100 g	A			✓	31	1
GLYCM	Glycogen (monosaccharide equivalents)	GLYCM	g/100 g	C			✓	33	1
HG	Mercury	HG	µg/100 g	A			✓	451	16
HIS	Histidine	HIS	mg/100 g	A			✓	593	21
HIS_G	Histidine (grams)	HIS_G	g/100 g	C			✓	592	21
HISN	Histidine (/g N)	HISN	mg/g N	C			✓	592	21
HYP	Hydroxyproline	HYP	mg/100 g	A			✓	41	2
HYP_G	Hydroxyproline (grams)	HYP_G	g/100 g	A			✓	41	2
HYPN	Hydroxyproline (/g N)	HYPN	mg/g N	C			✓	41	2
ID	Iodide	ID	µg/100 g	A	✓	✓	✓	2767	100
ILE	Isoleucine	ILE	mg/100 g	A			✓	593	21
ILE_G	Isoleucine (grams)	ILE_G	g/100 g	A			✓	592	21
ILEN	Isoleucine (/g N)	ILEN	mg/g N	C			✓	592	21
K	Potassium	K	mg/100 g	A	✓	✓	✓	2767	100
LACAC_G	Lactic acid (grams)	LACAC_G	g/100 g	A			✓	27	1
LACS	Lactose	LACS	g/100 g	A		✓	✓	2767	100
LACSM	Lactose (monosaccharide equivalents)	LACSM	g/100 g	C			✓	2767	100
LEU	Leucine	LEU	mg/100 g	A			✓	597	22
LEU_G	Leucine (grams)	LEU_G	g/100 g	A			✓	597	22
LEUN	Leucine (/g N)	LEUN	mg/g N	C			✓	596	22
LI	Lithium	LI	µg/100 g	A			✓	439	16
LUTN	Lutein	LUTN	µg/100 g	A			✓	87	3
LYCPN	Lycopene	LYCPN	µg/100 g	A			✓	132	5
LYS	Lysine	LYS	mg/100 g	A			✓	603	22
LYS_G	Lysine (grams)	LYS_G	g/100 g	A			✓	602	22
LYSN	Lysine (/g N)	LYSN	mg/g N	C			✓	602	22
MALAC_G	Malic acid (grams)	MALAC_G	g/100 g	A			✓	50	2

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
MALS	Maltose	MALS	g/100 g	A		✓	✓	2767	100
MALSM	Maltose (monosaccharide equivalents)	MALSM	g/100 g	C			✓	2767	100
MALTDEX	Maltodextrin		g/100 g	A			✓	28	1
MET	Methionine	MET	mg/100 g	A			✓	598	22
MET_G	Methionine (grams)	MET_G	g/100 g	A			✓	597	22
METN	Methionine (/g N)	METN	mg/g N	C			✓	597	22
MG	Magnesium	MG	mg/100 g	A		✓	✓	2767	100
MN	Manganese	MN	µg/100 g	A		✓	✓	2767	100
MNSAC	Monosaccharides, total	MNSAC	g/100 g	C			✓	2767	100
MO	Molybdenum	MO	µg/100 g	A			✓	556	20
NA	Sodium	NA	mg/100 g	A	✓	✓	✓	2767	100
NI	Nickel	NI	µg/100 g	A			✓	517	19
NIA	Niacin, preformed	NIA	mg/100 g	A		✓	✓	2767	100
NIAEQ	Niacin equivalents, total	NIAEQ	mg/100 g	C	✓	✓	✓	2767	100
NIATRP	Niacin equivalents from tryptophan	NIATRP	mg/100 g	C		✓	✓	2767	100
NT	Nitrogen, total	NT	g/100 g	A		✓	✓	2767	100
OXALAC_G	Oxalic acid (grams)	OXALAC_G	g/100 g	A			✓	2	<1
OA_G	Organic acids, total (grams)	OA_G	g/100 g	C			✓	119	4
P	Phosphorus	P	mg/100 g	A	✓	✓	✓	2767	100
PANTAC	Pantothenic acid	PANTAC	mg/100 g	A			✓	1598	58
PB	Lead	PB	µg/100 g	A			✓	656	24
PHE	Phenylalanine	PHE	mg/100 g	A			✓	604	22
PHE_G	Phenylalanine (grams)	PHE_G	g/100 g	A			✓	603	22
PHEN	Phenylalanine (/g N)	PHEN	mg/g N	C			✓	603	22
PHYSTR	Phytosterols, total	PHYSTR	mg/100 g	A			✓	35	1
PRO	Proline	PRO	mg/100 g	A			✓	578	21
PRO_G	Proline (grams)	PRO_G	g/100 g	A			✓	577	21
PRON	Proline (/g N)	PRON	mg/g N	C			✓	577	21
PROT	Protein, total; calculated from total nitrogen	PROT	g/100 g	C	✓	✓	✓	2767	100
PROXTOT	Proximate, total			C			✓	2767	100
PSACNS	Polysaccharides, non-starch	PSACNS	g/100 g	A			✓	2129	77
PSACNSI	Polysaccharides, non-starch, water-insoluble	PSACNSI	g/100 g	A			✓	1962	71

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
PSACNSS	Polysaccharides, non-starch, water-soluble	PSACNSS	g/100 g	C			✓	1962	71
QUINAC_G	Quinic acid (grams)	QUINAC_G	g/100 g	A			✓	5	<1
RB	Rubidium	RB	mg/100 g	A			✓	440	16
RETOL	Retinol	RETOL	µg/100 g	A		✓	✓	2767	100
RIBF	Riboflavin	RIBF	mg/100 g	A	✓	✓	✓	2767	100
S	Sulphur	S	mg/100 g	A			✓	1511	55
SE	Selenium	SE	µg/100 g	A	✓	✓	✓	2767	100
SER	Serine	SER	mg/100 g	A			✓	596	22
SER_G	Serine (grams)	SER_G	g/100 g	A			✓	595	22
SERN	Serine (/g N)	SERN	mg/g N	C			✓	595	22
SISOL	Silicon (acid soluble)	SISOL	µg/100 g	A			✓	300	11
SN	Tin	SN	µg/100 g	A			✓	530	19
STARCH	Starch, total	STARCH	g/100 g	A	✓	✓	✓	2767	100
STARCHM	Starch, total (monosaccharide equivalents)	STARCHM	g/100 g	C			✓	2767	100
STARES	Starch, resistant	STARES	g/100 g	A			✓	171	6
SUCAC_G	Succinic acid (grams)	SUCAC_G	g/100 g	A			✓	17	<1
SUCS	Sucrose	SUCS	g/100 g	A		✓	✓	2767	100
SUCSM	Sucrose (monosaccharide equivalents)	SUCSM	g/100 g	C			✓	2767	100
SUGAR	Sugars, total	SUGAR	g/100 g	C	✓	✓	✓	2767	100
SUGAD	Sugars, added	SUGAD	g/100 g	C	✓	✓	✓	2767	100
SUGFR	Sugars, free		g/100 g	C	✓	✓	✓	2767	100
SUGARM	Sugars, total (monosaccharide equivalents)	SUGARM	g/100 g	C			✓	2767	100
TAU	Taurine	TAU	mg/100 g	A			✓	160	6
TAU_G	Taurine (grams)	TAU_G	g/100 g	A			✓	160	6
TAUN	Taurine (/g N)	TAUN	mg/g N	C			✓	160	6
THIA	Thiamin	THIA	mg/100 g	A	✓	✓	✓	2767	100
THR	Threonine	THR	mg/100 g	A			✓	597	22
THR_G	Threonine (grams)	THR_G	g/100 g	A			✓	596	22
THRN	Threonine (/g N)	THRN	mg/g N	C			✓	596	22
TOCPHA	Alpha-tocopherol	TOCPHA	mg/100 g	A		✓	✓	1850	67
TOCPHB	Beta-tocopherol	TOCPHB	mg/100 g	A		✓	✓	678	24
TOCPHBG	Beta-tocopherol + Gamma-tocopherol		mg/100 g	A			✓	6	<1

Component identifier	Component name	INFOODS Tagname	Unit	Analysed (A) or Calculated (C)	Concise Table 13 th Edition 2018	FOODfiles™ 2018 Version 01			
						Standard	Unabridged	Number of foods	Percent
TOCPHD	Delta-tocopherol	TOCPHD	mg/100 g	A		✓	✓	1060	38
TOCPHG	Gamma-tocopherol	TOCPHG	mg/100 g	A		✓	✓	1095	40
TRP	Tryptophan	TRP	mg/100 g	A or C		✓	✓	1422	51
TRP_G	Tryptophan (grams)	TRP_G	g/100 g	A			✓	736	27
TRPN	Tryptophan (/g N)	TRPN	mg/g N	C			✓	661	24
TYR	Tyrosine	TYR	mg/100 g	A			✓	603	22
TYR_G	Tyrosine (grams)	TYR_G	g/100 g	A			✓	602	22
TYRN	Tyrosine (/g N)	TYRN	mg/g N	C			✓	602	22
V	Vanadium	V	µg/100 g	A			✓	407	15
VAL	Valine	VAL	mg/100 g	A			✓	597	22
VAL_G	Valine (grams)	VAL_G	g/100 g	A			✓	596	22
VALN	Valine (/g N)	VALN	mg/g N	C			✓	596	22
VITA	Vitamin A, retinol equivalents	VITA	µg/100 g	C	✓	✓	✓	2767	100
VITA_RAE	Vitamin A, retinol activity equivalents	VITA_RAE	µg/100 g	C		✓	✓	2767	100
VITB12	Vitamin B12	VITB12	µg/100 g	A	✓	✓	✓	2767	100
VITB6A	Vitamin B6	VITB6A	mg/100 g	A	✓	✓	✓	2767	100
VITC	Vitamin C	VITC	mg/100 g	A	✓	✓	✓	2767	100
VITD	Vitamin D; calculated by summation	VITD	µg/100 g	C	✓	✓	✓	2767	100
VITE	Vitamin E, alpha-tocopherol equivalents	VITE	mg/100 g	C	✓	✓	✓	2767	100
VITK	Vitamin K	VITK	µg/100 g	A			✓	142	5
WATER	Water	WATER	g/100 g	A	✓	✓	✓	2767	100
XCHGC	Carbohydrate exchanges		Exchanges/100 g	C			✓	2767	100
XFA	Total fat-to-total fatty acids conversion factor	XFA	g/g				✓	2175	79
XN	Nitrogen-to-protein conversion factor	XN	g/g				✓	2767	100
ZEAX	Zeaxanthin	ZEAX	µg/100 g	A			✓	88	3
ZN	Zinc	ZN	mg/100 g	A	✓	✓	✓	2767	100

Appendix 2. Source codes

This table provides the key for the lower-case letter codes that indicate the source of data, such as food composition analysis, calculation, or borrowed from another database.

Source Code	Source	Reference
a	Australian Food Composition Database (previously called NUTTAB)	Food Standards Australia New Zealand (2019)
b	UK/National Nutrient Databank/McCance and Widdowson's The Composition of Foods Integrated Dataset (CoFID)	Holland et al. (1991); Food Standards Agency (2002b); Finglas et al. (2015b)
c	Calculated by FIMS after data entry	Calculated by internally developed Java application
d	Derived from published sources	Finglas et al. (2015a); Food and Agriculture Organisation of the United Nations and World Food Organisation of the United Nations. (2002); Greenfield & Southgate (2003)
f	Fortified	Manufactured Food Database (2011)
g	Data estimated based on the ingredients or related foods	
h, s, j	Pacific Islands Food Composition Tables and publications	Bradbury & Holloway (1988); English et al. (1996); Dignan et al. (2004)
l	Analytical value less than the limit of detection/quantitation	
m	Manufacturers supplied data	Nutrition Information Panel (NIP)
n	Papua New Guinea	
o	Other overseas Databases:	
	Health Canada – Canadian Nutrient File (CNF)	Health Canada (2015)
	Danish Food Composition Databank	Saxholt et al. (2009)
	Finnish Food Composition Database-FINELI	National Institute for Health and Welfare. (2010)
	Food Composition database for Epidemiological Studies in Italy	Gnagnarella & Salvini (2008)
	Health Promotion Board-Nutrient Composition of Foods (Singapore)	Health Promotion Board (2003)
p	Presumed zero	Based on the published sources, food is not a source for components
r	Value derived from a related food (imputed)	
u	USDA National Nutrient Database	U.S. Department of Agriculture (2015)
v	Food Composition Table for Use in Africa	Wu Leung (1968)
w	Value derived from sample with unknown dry matter/water; may not be directly related to water value presented	
x	No confidence in selected value	
y	Australian Aboriginal Foods	Miller et al. (1993)
z	New Zealand analytical data	

Appendix 3. Analytical methods

The following table provides the details of the analytical methods used to generate the data for components in the database.

Components	Method	Method identification and references ^a	Lower limit of detection *
Alcohol	Gas chromatography	Palo & Ilkova (1970)	
Ash	Gravimetric	AOAC 900.02: Ash of sugars and syrups; AOAC 920.153: Ash of meat AOAC 923.03: Ash of flour AOAC 938.08: Ash of seafood AOAC 940.26: Ash of fruits and fruit products AOAC 942.05: Ash in animal feed In-house method	0.1 g/100 g
Fat, total	Acid hydrolysis – Organic solvent extraction/Soxhlet extraction	AOAC 954.02: Fat (crude) or ether extract in pet food. AOAC 948.22: Fat (crude) in nuts and nut products	0.1 g/100 g
	Alkaline hydrolysis - Organic solvent extraction	International Dairy Federation (2004c)	unknown
	Alkaline and acid hydrolysis - Organic solvent extraction	In-house method	unknown
Nitrogen, total	Combustion/Kjeldahl/mixed catalyst	AOAC 988.05: Protein (crude) in animal feed and pet food AOAC 968.06: Protein (crude) in animal feed AOAC 978.04: Nitrogen (total) (crude protein) in plant AOAC 981.10: Crude protein in meat AOAC 991.20: Nitrogen (total) in milk	0.1 g/100 g
Fatty acid profile	Methylated with acid catalyst - Gas chromatography	Bannon et al. (1985); Sukhija & Palmquist (1988)	0.1 g/100 g
Conjugated fatty acids	Methylated with base catalyst - Gas chromatography	In house method based on Aldai et al. (2007)	0.1 g/100 g
Fibre, total dietary	Enzymatic-gravimetric	AOAC 991.43:	0.1 g/100 g
Polysaccharides, non-starch	Enzymatic-gravimetric	Englyst et al. (1988); Englyst et al. (1994)	0.1 g/100 g
Sugar profile	Gas liquid chromatography	In-house method	0.1 g/100 g
Starch	Enzymatic-colometric	AOAC 996.11: Starch (total) in cereal products	0.1g/100 g
Starch, resistant	Enzymatic-colometric	AOAC 2002.2: Resistant starch in starch and plant materials	0.1 g/100 g
Water/Total solid	Vacuum oven/forced air-gravimetric	AOAC 950.46: Moisture in meat AOAC 930.15: Loss on drying (moisture) for feeds (at 135°C for 2 hours)/dry	0.1 g/100 g

Components	Method	Method identification and references ^a	Lower limit of detection *
		matter on oven drying for feeds (at 135°C for 2 hours) AOAC 925.10: Solids (total) and loss on drying (moisture) in flour AOAC 925.45: Loss on drying (moisture) in sugars International Dairy Federation (2010) International Dairy Federation (2004a) International Dairy Federation (2004b)	
Ca, K, Mg, Na, P	Biological materials digestion, ICP-OES	AOAC 984.27: Calcium, copper, iron, magnesium, manganese, phosphorous and zinc in infant formula.	0.001–0.005 g/100 g
Fe		Mendham et al. (2000) Martin et al. (1994)	1–5 mg/kg
Al, As, B, Co, Cr, Cs, Cu, Hg, Li, Mn, Mo, Pb, Ni, Rb, Sn, V, Zn, Cd	Biological materials digestion, ICP-MS	AOAC 984.27: Calcium, copper, iron, magnesium, manganese, phosphorous and zinc in infant formula. Mendham et al. (2000) Martin et al. (1994)	unknown
Cl	Potentiometric	AOAC 971.27: Sodium chloride in canned vegetables	10 mg/kg dry or fresh product, 1 mg/kg liquids
F	Specific Ion electrode	Dabeka et al. (1979)	
I, Se, As	TMAH digestion, ICP-MS	Fecher et al. (1998)	0.001–0.004 mg/kg
Alpha- and Beta-carotene	HPLC	European Committee for Standardisation (2000b)	5 µg/100 g
Lutein and Zeaxanthin	HPLC	McGhie & Ainge (2002) McGhie & Rowan (2012)	unknown
Retinol	Colorimetric/HPLC	AOAC 974.29: Vitamin A in mixed feeds, premixes, and human and pet foods Brubacher et al. (1985b); European Committee for Standardisation (2000a)	10 IU/100 g
Folate	Microbiological	DeVries et al. (2005)	2.75 µg/100 g
Thiamin	HPLC-fluorometric	European Committee for Standardisation (2003a)	0.01 mg/100 g
Riboflavin	Fluorometric	Egberg & Potter (1975) Dunbar & Stevenson (1979)	0.01 mg/100 g
Niacin, preformed	HPLC	Woollard (1984)	0.01 mg/100 g
Pantothenic acid (Vitamin B ₅)	Gas liquid chromatography	Davidek et al. (1985)	1 mg/100 g
Vitamin B ₆	HPLC-fluorometric detection	Reitzer-Bergaentzle et al. (1993) Bitsch & Moller (1989) Olds et al. (1993)	0.01 mg/100 g
Vitamin B ₁₂	Radio isotope dilution	In-house HPLC, total as cyanocobalamin	0.10 µg/100 g
Vitamin C	LC-fluorometric detection	Dodson et al. (1992)	1 mg/100 g

Components	Method	Method identification and references ^a	Lower limit of detection *
		European Committee for Standardisation (2003b)	
25-hydroxyvitamin D ₃	Acetonitrile extraction followed by Radioimmune assay	Diasorin kit (Stillwater, Minnesota) from Immuno Pty Ltd	unknown
Cholecalciferol (vitamin D ₂) & Ergocalciferol (vitamin D ₃)	Reversed phase HPLC with UV detection	AOAC 2002.05: Cholecalciferol (Vitamin D ₃) in selected foods	1 µg/100 g
Vitamin K1	HPLC	Damon et al. (2005)	unknown
Alpha-Beta-, Delta- and Gamma-tocopherol	HPLC	Brubacher et al. (1985a); AOAC 971.27	0.11 IU/100 g
Biotin	Optical biosensor-based immunoassay	Indyk et al. (2000)	unknown
Caffeine	HPLC	Woollard (1982)	1 mg/100 g
Cholesterol	Gas chromatography	AOAC 933.08: Residue (unsaponifiable) of oils and fats AOAC 970.50: Fat (vegetable) in butterfat AOAC 970.51: Fats (animal) in vegetable fats and oils (determination of cholesterol)	0.5 mg/100 g
Amino acids	HPLC	AOAC 988.15: Tryptophan in foods and food and feed ingredients AOAC 985.28: Sulphur amino acids in food, feed ingredient, and processed foods AOAC 994.12: Amino acids in feeds (Landry et al. 1992)	unknown
Organic acids	HPLC	Doyon et al. (1991)	unknown

^a AOAC methods source: Association of Official Analytical Chemists (2005).

* Lower limit of detection varies based on the food matrix.

Note: The food components are analysed by IANZ (International Accreditation New Zealand) accredited laboratories in New Zealand and NATA (The National Association of Testing Authorities, Australia) accredited laboratory in Australia. Some of the analytical methods were not accredited for some of the food matrixes.

Appendix 4. Examples for energy calculation

1. Energy calculated based on FAO/INFOODS (2012a)

Component name: energy, total metabolisable (kJ)	
Components Identifier	ENERC
INFOODS tagname	ENERC
Formula	Energy (kJ/100 g Edible Portion, EP) = (protein, total calculated from nitrogen total x 17) + (fat, total x 37) + (available carbohydrate by weight x 17) + (alcohol x 29)
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains per g 100 g EP: 1.023 g protein, total; calculated from nitrogen total (PROT) + 0.28 g fat, total (FAT) + 12.42 g available carbohydrate by weight (CHOAVL) + 0.00 g alcohol (ALC) Calculation: [(1.023 g PROT x 17) + (0.28 g FAT x 37) + (12.42 g CHOAVL x 17) + (0.00 g ALC x 29)] = ENERC = 238.90 kJ/100 g EP

Component name: energy, total metabolisable (kcal)	
Components Identifier	ENERC_KCAL
INFOODS tagname	ENERC_KCAL
Formula	Energy (kcal/100 g EP) = (protein, total; calculated from nitrogen total x 4) + (fat, total x 9) + (available carbohydrate by weight x 4) + (alcohol x 7)
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains per g 100 g EP: 1.023 g protein, total; calculated from nitrogen total (PROT) + 0.28 g fat, total (FAT) + 12.42 g available carbohydrate by weight (CHOAVL) + 0.00 g alcohol (ALC) Calculation: [(1.023 g PROT x 4) + (0.28 g FAT x 9) + (12.42 g CHOAVL x 4) + (0.00 g ALC x 7)] = ENERC_KCAL = 56.29 kcal/100 g EP

2. Energy calculated according to FAO/INFOODS (2012a) including dietary fibre

Component name: energy, total metabolisable (including dietary fibre, kJ)

Components Identifier	ENERC1
INFOODS tagname	None
Formula	Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 17) + (fat, total x 37) + available carbohydrate by weight (CHOAVL) x 17) + (alcohol x 29) + (fibre, total dietary x 8)
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains per g 100 g EP: 1.023 g protein, total; calculated from nitrogen total (PROT) + 0.28 g fat, total (FAT) + 12.42 g available carbohydrate by weight (CHOAVL) + 0.00 g alcohol (ALC) + 1.43 g fibre, total dietary (FIBTG) Calculation: [(1.023 g PROT x 17) + (0.28 g FAT x 37) + (12.42 g CHOAVL x 17) + (0.00 g ALC x 29) + (1.43 g FIBTG x 8)] = ENERC1 = 250.33 kJ/100g EP

Component name: energy, total metabolisable (including dietary fibre, kcal)

Components Identifier	ENERC1_KCAL
INFOODS tagname	None
Formula	Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 4) + (fat, total x 9) + available carbohydrate by weight (CHOAVL) x 4) + (alcohol x 7) + (fibre, total dietary x 2)
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains per g 100 g EP: 1.023 g protein, total; calculated from nitrogen total (PROT) + 0.28 g fat, total (FAT) + 12.42 g available carbohydrate by weight (CHOAVL) + 0.00 g alcohol (ALC) + 1.43 g fibre, total dietary (FIBTG) Calculation: [(1.023 g PROT x 4) + (0.28 g FAT x 9) + (12.42 g CHOAVL x 4) + (0.00 g ALC x 7) + (1.43 g FIBTG x 2)] = ENERC1_KCAL = 59.15 kcal/100 g EP

3. Energy calculated according to Food Standards Australia New Zealand (2018), using carbohydrate by difference

Component name: energy, total metabolisable, carbohydrate by difference, FSANZ (kJ)

Components Identifier	ENERC_FSANZ1
INFOODS tagname	N/A
Formula	Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 17) + (fat, total x 37) + (carbohydrate by difference x 17) + (alcohol x 29) + (fibre, total dietary x 8) + (organic acids, total x 13)
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains per g 100 g EP: 1.023 g protein, total; calculated from nitrogen total (PROT) + 0.28 g fat, total (FAT) + 14.36 g carbohydrate by difference (CHOAVLDF_FSANZ) + 0.00 g alcohol (ALC) + 1.43 g fibre, total dietary (FIBTG) + 2.25 g organic acid, total (gram) (OA_G) Calculation: [(1.023 g PROT x 17) + (0.28 g FAT x 37) + (12.11 g CHOAVLDF_FSANZ x 17) + (0.00 g ALC x 29) + (1.43 g FIBTG x 8) + (2.25 g OA_G x 13)] = ENERC_FSANZ1 = 274.28 kJ/100 g EP

Component name: energy, total metabolisable, carbohydrate by difference, FSANZ (kcal)

Components Identifier	ENERC_FSANZ1_KCAL
INFOODS tagname	N/A
Formula	Energy (kcal/100 g EP) = Energy, total metabolisable, carbohydrate by difference FSANZ (kJ/100 g EP)/4.18
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains 274.28 kJ per 100 g EP of energy, total metabolisable, carbohydrate by difference, FSANZ (kJ) (ENERC_FSANZ1) Calculation: [(274.28 kJ/100 g EP ENERC_FSANZ1)/ 4.18] = ENERC_FSANZ1_KCAL = 65.62 kcal/100 g EP

4. Energy calculated according to Food Standards Australia New Zealand (2018), using available carbohydrate

Component name: energy, total metabolisable, available carbohydrate, FSANZ (kJ)

Components Identifier	ENERC_FSANZ2
INFOODS tagname	None
Formula	Energy (kJ/100 g EP) = (protein, total; calculated from nitrogen total x 17) + (fat, total x 37) + (available carbohydrate x 17) + (alcohol x 29) + (fibre, total dietary x 8) + (organic acids, total x 13)
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains per g 100 g EP: 1.023 g protein, total; calculated from nitrogen total (PROT) + 0.28 g fat, total (FAT) + 12.42 g available carbohydrate (CHOAVL_FSANZ) + 0.00 g alcohol (ALC) + 1.43 g fibre, total dietary (FIBTG) + 2.25 g organic acids, total (OA_G) Calculation: [(1.023 g PROT x 17) + (0.28 g FAT x 37) + (12.42 g CHOAVL_FSANZ x 17) + (0.00 g ALC x 29) + (1.43 g FIBTG x 8) + (2.25 g OA_G x 13)] = ENERC_FSANZ2 = 279.52 kJ/100 g EP

Component name: energy, total metabolisable, available carbohydrate, FSANZ (kcal)

Components Identifier	ENERC_FSANZ2_KCAL
INFOODS tagname	None
Formula	Energy (kcal/100 g EP) = Energy, total metabolisable, available carbohydrate, FSANZ (kJ) (kJ/100 g EP)/4.18
Example	L1077, Kiwifruit, gold, flesh & seed, fresh, raw, Zespri® SunGold Kiwifruit contains 279.52 kJ/100 g EP of energy, total metabolisable, available carbohydrate, FSANZ (kJ) (ENERC_FSANZ2) Calculation: [(279.52 kJ/100 g EP ENERC_FSANZ2)/ 4.18] = ENERC_FSANZ2_KCAL = 66.87 kcal/100 g EP

Appendix 5. Method for estimating added and free sugars

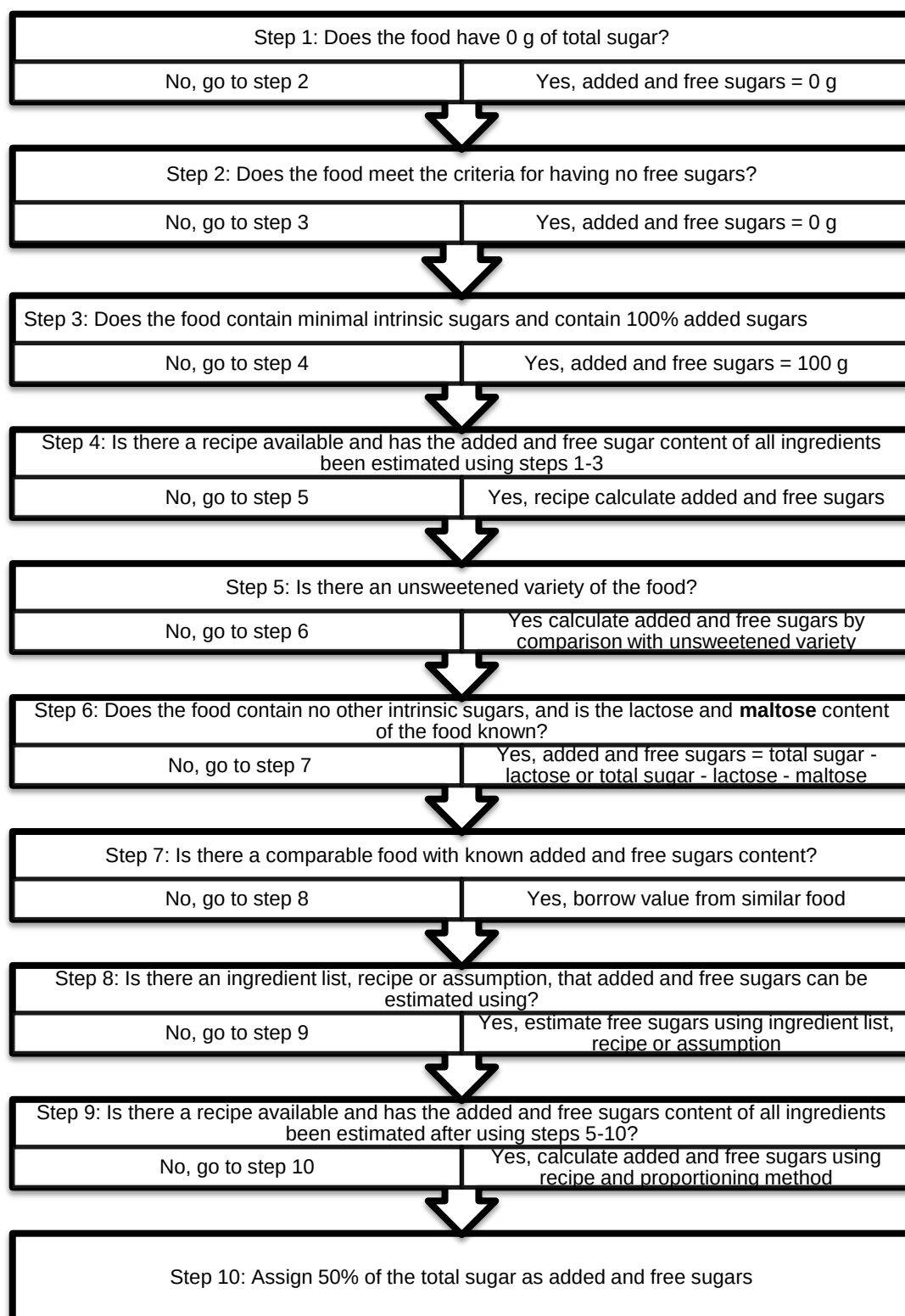


Figure 1. Ten-step method for estimating added and free sugars content (Louie et al. 2015)

Detailed 10-Step method for estimating added and free sugars

Steps	Added sugar	Free sugar
Objective steps		
1	All foods with the total sugar content of 0 g were assigned 0 g sugar	
	Assigned 0 g added sugar to foods in the following groups	Assigned 0 g free sugar to foods in the following groups
2	(a) 100% fruit juice and vegetable juice with no added sugar (include fruit puree) (b) All spices and herbs (c) All fats and oils (d) Plain cereals including pastas, rice and flours (e) Eggs and egg products (excluding egg-based desserts) (f) Fresh fruit, vegetables (including salads with no dressing), fresh meat, fresh sea food and tofu. Include cooked vegetables, meat and sea foods with no added sugar. (g) Fruit canned in 100% fruit juice or liquid sweetened with artificial sweeteners (h) Beverage base (without added sugar) and prepared with water (including flavoured drink) (i) Legumes (fresh, dried and/or processed, except sweetened varieties) (j) Mixed dishes with no added sugar (decided on the basis of ingredient information; recipe) (k) Non-sugar-sweetened alcoholic beverage (l) Non-sugar-sweetened tea and coffee (m) Non-sugar-sweetened milk (including plant-based milk) and buttermilk; breastmilk	(a) See below step 3 (l) (b) All spices and herbs (c) All fats and oils (d) Plain cereals including pastas, rice and flours (e) Eggs and egg products (excluding egg-based desserts) (f) Fresh fruit, vegetables (including salads with no dressing), fresh meat, fresh sea food and tofu. Include cooked vegetables, meat and sea foods with no added sugar. (g) See below step 3 (l) (h) Beverage base (without added sugar) (i) Legumes (fresh, dried and/or processed, except sweetened varieties) (j) Mixed dishes with no added sugar (decided on the basis of ingredient information; recipe) (k) Non-sweetened alcohol (excluding liqueurs and mixers) (l) Non-sugar-sweetened tea and coffee (m) Non-sugar-sweetened milk (excluding plant-based milk) and buttermilk; breastmilk

Steps	Added sugar	Free sugar
	(n) Non-sweetened dairy products (including those sweetened with artificial sweeteners only) (o) All nuts and seeds (does not include sweetened varieties or nut bars) (p) Oats, including porridge with no added sugar (q) Plain pastries without filling (such as chocolate, dried fruits and/or nuts) (r) Plain breads (except gluten-free), English muffin, bagels, pizza bases and naan (s) Unsweetened dried fruits	(n) Non-sweetened dairy products (including those sweetened with artificial sweeteners only) (o) All nuts and seeds (does not include sweetened varieties or nut bars) (p) Oats, including porridge with no added sugar (q) Plain pastries without filling (such as chocolate, dried fruits and/or nuts) (r) Plain breads (except gluten-free), English muffin, bagels, pizza bases and naan (s) Unsweetened dried fruits
3	Assigned 100% of total sugar as added sugar to foods in the following groups (a) All confectionary except those containing dairy products such as fudge and chocolate. Vegetable chips (including potato, cassava, soy etc.) (b) Breakfast cereals and cereal bars without fruits, chocolate, dairy or milk (c) Coffee and beverage base with no milk solids, dry or made up with water (d) Crumbed/battered meat and seafood (e) Processed meat including pies, pastries, sausages etc. (f) Regular soft drinks, sport drinks, flavoured water and non-fruit-based products that do not contain fruit, chocolate or dairy products (g) Savoury biscuits, sweet biscuits, cakes, and buns, donuts and batter-based products that do not contain fruit, chocolate or dairy products (h) Soy beverages and soy yoghurt without added fruits (i) Stock powder	Assigned 100% of total sugar as added sugar to foods in the following groups (a) All confectionary (excluding chocolate) and vegetable chips (including potato, cassava, soy etc.) (b) All breakfast cereals and cereal bars which do not contain fruit pieces, chocolate, or dairy or milk (c) Coffee and beverage bases (d) Crumbed/battered meat and seafood (e) Processed meat including pies, pastries, sausages etc. (f) Sugar-sweetened soft drinks, sports drinks, flavoured waters, and energy drinks. (g) All baked goods such as biscuits, cakes, buns, donuts and crackers that did not contain fruits, chocolate or dairy products. Gluten free breads. (h) Soy beverages and soy yoghurt without added fruits (i) Stock powder

Steps	Added sugar	Free sugar
	(j) Sugars and syrups	(j) All sugars and syrups
	(k) Sauces and dressing, excluding pasta sauces and those that are vegetable-based such as pickles	(k) Sauces and dressing, excluding pasta sauces and those that are vegetable-based such as pickles
		(l) All fruit juices, purees, concentrates, fruit canned in fruit juice and jams (both sweetened and unsweetened varieties), including tomato pastes, sauces, and purees
		(m) Alcoholic mixed-drink, Non-cream-based liqueurs
4	Calculation based on standard recipe used in the food composition database where added sugar or free sugar contents of all ingredients were available from the steps 1–3.	
	(a) Calculation based on standard recipe used in the food composition database (by recipe calculation by FIMS)	
	(b) Added sugar content determined for the undrained canned fruits in syrup: Added sugar (g/100 g) = Total sugar (g/100 g) – (total sugar content of raw fruit (g/100 g) * % of raw fruit in the canned fruits.	Free sugar content determined for the undrained canned fruits in fruit juice or syrup: Free sugar (g/100 g) = Total sugar (g/100 g) – (total sugar content of raw fruit (g/100 g) * % of raw fruit in the canned fruits.
5	Calculation based on comparison with values from the unsweetened variety	
	Added or free sugar per 100 g is given by the formula:	
	$\text{Added or Free sugar (g/100g)} = \frac{100 \times (S_{US} - S_T)}{(S_{US} - 100)}$	
	Where S_{US} is the total sugar content per 100 g of the unsweetened variety of the food and S_T is the total sugar for the food item that free sugars to be estimated.	
	<i>Example: Added and free sugars calculation for sweetened (L1124) variety from the unsweetened (L1128) counterpart</i>	
	<i>L1124 - Peach, in light syrup, sugar-sweetened, canned, drained, composite (sugar total 14.2 g/100 g)</i>	
	<i>L1128 - Peach, in water, artificially-sweetened, canned, drained, composite (sugar total 6.6 g/100 g)</i>	
	<i>Added or free sugar for L1124 = $100 - (6.6 - 14.2) / (6.6 - 100) = 8.14 \text{ g/100 g}$</i>	

Steps	Added sugar	Free sugar
6	Decision based on analytical data of mono- and disaccharide content	
	Lactose subtracted from total sugars for all foods Maltose subtracted from total sugars as intrinsic sugar	
7	Use borrowed values from similar products from steps 1–6 or from overseas databases	
	Example H215 - Spring roll, vegetarian, deep-fried, commercial, added and free sugar estimated based on the recipe in the NZFCD (old archive Food Records, H130 - Spring roll, deep), 90% of the total sugar.	
8	Subjective estimation of added and free sugars based on the best available information regarding ingredients and/or common recipes and/or assumption.	
	Added and free sugar estimated by searching the ingredient list of packed foods and recipes for sources. Example: D1055 – Breakfast cereal, muesli, ready to eat, Toasted Muesli Super Fruity, Sanitarium, fortified vitamin E Ingredient list states dried fruit = 16% [raisins, cranberries]. Assumed even distribution (8.0% each). Total sugar from 8.0% raisins, 8.0% cranberries were considered as intrinsic sugar and deducted from the total sugar of D1055.	
9	Calculate based on standard recipe which include ingredients with values assigned at steps 5–8, using the proportioning method. Step 4 is repeated here where more have their added sugar contents estimated after steps 5–8.	
	Ingredients ratio in the standard recipe checked (step 8, % dried fruit), exclude lactose value (step 6). Example – A1152 - Cake, assorted fruits, rich or dark, uniced, ready to eat, commercial	
10	Assign 50% of total sugars as added or free sugars	
	Used when it was not possible to determine free sugars using steps 1–9. Example takeaway foods, H1058 - Salad, coleslaw with dressing, fresh, from deli, composite, assumed 50% of the total is added and free sugars.	

Sources: Louie et al. (2015), (Louie et al. 2016) and (Kibblewhite et al. 2017).

Appendix 6. Individual fatty acids contributing to the total fatty acids

Component identifier										
Individual fatty acids	Undifferentiated fatty acid	FASATF	FAMSF	FAPUF	FAPUN3F	FALCPUN3F	FAPUN6F	FAMSTF	FAPUTF	FATRNF
F4D0F		✓								
F6D0F		✓								
F8D0F		✓								
F10D0F		✓								
F12D0F		✓								
F13D0F		✓								
F14D0F		✓								
F15D0F		✓								
F16D0F		✓								
F17D0F		✓								
F18D0F		✓								
F20D0F		✓								
F21D0F		✓								
F22D0F		✓								
F23D0F		✓								
F24D0F		✓								
F10D1F			✓							
F12D1F			✓							
F14D1N5F	F14D1F		✓							
F15D1F			✓							
F16D1CF			✓							
F16D1TF	F16D1F		✓					✓		✓
F17D1F			✓							
F18D1CN7F										
F18D1TN7F								✓		✓
F18D1CN9F	F18D1F		✓							
F18D1TN9F								✓		✓
F20D1N9F										
F20D1N11F	F20D1F		✓							

Component identifier										
Individual fatty acids	Undifferentiated fatty acid	FASATF	FAMSF	FAPUF	FAPUN3F	FALCPUN3F	FAPUN6F	FAMSTF	FAPUTF	FATRNF
F22D1N9F	F22D1		✓							
F22D1N11F										
F24D1F			✓							
F18D2CN6F	F18D2						✓			
F18D2TN6F				✓					✓	
F18D2TCN9T11F									✓	
F18D3N3F	F18D3F				✓					
F18D3N6F				✓			✓			
F18D4N3F	F18D4			✓	v					
F20D2N6F	F20D2			✓			✓			
F20D3N3F	F20D3				✓	✓				
F20D3N6F				✓			✓			
F20D4N3F	F20D4				✓	✓				
F20D4N6F				✓			✓			
F20D5N3F	F20D5			✓	✓	✓				
F21D5N3F	F21D5			✓	✓	✓				
F22D2N6F	F22D2			✓			✓			
F22D4N6F	F22D4			✓			✓			
F22D5N3F	F22D5				✓	✓				
F22D5N6F				✓			✓			
F22D6N3F	F22D6F			✓	✓	✓				

Appendix 7. Example for calculation of Yield Factors

An example calculation for a recipe with a single ingredient that has been cooked N1021 (sausage, beef, pan-fried, with no added fat, composite) of True WYF, Water YF and Fat YF.

Calculation of WYF

Weight of the ingredient (sausage raw) (by weighing) – 234.0 g

Weight of the cooked dish (by weighing) – 190.2 g

$$WYF (\%) (Equation 34) = \frac{190.2 (g)}{234.0 (g)} \times 100 = 81.3 (\%)$$

Calculation of Water YF

Water content in total ingredients (by analysis) – 59.6g/100 g

Water content in cooked dish (by analysis) – 56.7g/100 g

$$Water YF (\%) (Equation 35) = \frac{56.7 (g/100 g)}{59.6 (g/100 g)} \times 81.3 (\%) = 77.3 (\%)$$

Calculation of Fat YF

Fat content total ingredients (by analysis) – 20.4 g/100 g

Fat content in cooked dish (by analysis) – 19.3 g/100 g

$$Fat YF (\%) (Equation 36) = \frac{19.3 (g/100 g)}{20.4 (g/100 g)} \times 81.3 (\%) = 76.9 (\%)$$

Appendix 8. Recipe calculation – Mixed method

Examples showing nutrient calculation for a recipe food record to demonstrate the principles of the recipe calculation system 'Mixed method'

Note: The FIMS algorithm used to calculate macro and micro nutrient values in recipes is very complex; the manual calculation below has been simplified by excluding the weight changes from other than water and fat. As a result the manually calculated nutrient values are not an exact match with FIMS calculated values in the data files. These examples are intended to demonstrate how the recipe calculation system 'Mixed method' works and the values appearing in FOODfiles 2018 V 01 are derived.

Single ingredient recipe: N1021 – Sausage, beef, pan-fried, with no added fat, composite.

Ingredient	Amount in recipe g	Water g	Fat g	Niacin mg	Sodium mg	Iron mg	Retinol µg
N1015 - Sausage, beef, fresh, raw, composite (100 g)	234 (g)	59.60	20.40	2.13	630	1.39	16.5
Total food components in the recipe – (A)	234 (X)	139.46	47.74	4.98	1474.2	3.25	38.61
YF and NRF (g/g) – (B)		0.773 ^a	0.769 ^a	0.75 ^b	0.75 ^b	0.65 ^b	0.75 ^b
Total nutrients in cooked dish = (A x B)		107.81	36.71	3.74	1105.65	2.11	28.96
Weight loss = (A - (A x B))		31.66 (Y)	11.03 (Z)				
Weight of the cooked dish	191.32 = (X - (Y+Z)) ^c						
Total food component in cooked dish (100 g) (Equation 38)		56.35	19.19	1.95	577.92	1.11	15.14

^a Yield Factors (YFs) Water and Fat were estimated from the analytical values (Appendix 7).

^b Nutrient Retention Factors (NRF) are imputed from related foods referenced in the U.S. Department of Agriculture (2007).

^c Weight changes of micro nutrients niacin, sodium, iron etc. during cooking are not included in the manual calculation above but are included in the recipe calculation by FIMS.

Multiple ingredients recipe: R5424 Pudding, rice cocoa & coconut, cooked.

Ingredient	Amount in recipe ^a	Weight ratio (X)	Water g	Fat g	Sodium mg	Iron µg	Folate, total µg
C1036 - Baking cocoa, cocoa, powder, Cocoa, Bournville, Cadbury (per 100 g)	37.75 g	0.02	2.7	15.8	102	9.8	65
E28 - Rice, white, polished, raw (per 100 g) – (A)	206 g	0.11	11.7	1.00	6	0.48	20
NRF ^b - (B)					1	0.95	0.7
E28 - Rice, white, polished, raw, (per 100 g), nutrients values adjusted with NRF = (A × B)					6	0.45	14
P10 - Salt, table, uniodised (per 100 g)	5.4 g	0.01	0.1	0	38100	0.2	0
Q1015 - Coconut, milk, standard, composite (per 100 g)	452.4 g	0.24	74.25	21.05	18.7	0.62	0
C40 - Water, tap (per 100 g)	1000 g	0.52	99.99	0	0.98	0	0
W23 - Sugar, raw (per 100 g)	218 g	0.11	0.3	0	1	0.3	0
Food components in the raw recipe (per 100 g) -(weight sum of nutrient in each ingredient *X)			70.93	5.38	114.86	0.42	65.73
Total weight of the ingredients	1919.55 g (Y)						
Food components in the raw recipe [(weight sum of nutrient in each ingredient (per 100 g)*X)*Y/100]			1361.59	103.25	2204.84	8.15	65.74
Yield Factors (per g) ^c			0.96	1			
Moisture and fat content in cooked dish			1307.12 ^d	103.25			
Water and fat loss (equal to total weight loss)			54.46 (Z)	0 (Z')			
Weight of the cooked dish	1865.09 =(Y-(Z+Z'))		1307.12	103.25 ^d	2204.84 ^d	8.10 ^d	53.38 ^d
Food components in cooked dish (per 100 g, R5424) (Equation 36)			70.08	5.53	118.22	0.43	2.86

Appendix 9. Key to abbreviations

Abbreviation	Meaning
ABV	Alcohol By Volume
AP	Applix Presents file extension
ASCII	American Standard Code for information changes
AOAC	Association of Official Analytical Chemists
CoFID	Composition of Foods Integrated Dataset
CNF	Canadian Nutrient Files
CSM	Common Standard Measure
FAO	Food and Agriculture Organisation (United Nations)
FIMS	Food Information Management System
FT	Full Text index file extension
FSANZ	Food Standard Australia and New Zealand
HPLC	High Performance Liquid Chromatography
ICP-MS	Inductivity Coupled Plasma Mass Spectrometry
ICP-OES	Inductivity Coupled Plasma Atomic Emission Spectroscopy
IUPAC	International Union of Pure and Applied Chemistry
INFOODS	International Network of Food Data Systems
LC	Liquid Chromatography
NIP	Nutrition Information Panel
NRF	Nutrient Retention Factor
NUTTAB	Nutrition Tables
NZFCDB	New Zealand Food Composition Database
PFR	Plant & Food Research
RAE	Retinol Activity Equivalent
RE	Retinol Equivalent
TFA	Total Fatty Acid
TMAH	Tetramethylammonium hydroxide
USDA	United States Department of Agriculture
YF	Yield Factor
WYF	Weight Yield Factor

Appendix 10. Units of measurements and New Zealand metric standards

Units	Definition
1 teaspoon	5 mL
1 tablespoon	15 mL
1 cup	250 mL
mL	millilitres
l	litre
cm ³	cubic centimetre
µg	microgram
mg	milligram
g	grams
kg	kilogram
cm	centimetre
kcal	kilocalories
kJ	kilojoules
g/100 g TFA	gram per 100 g of total fatty acids or gram per percentage of total fatty acids
mg/g N	milligram amino acid per gram nitrogen present in food
°C	degree Celsius
%ABV	Percentage alcohol by volume



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